Gauge-Covariant Gaussian Attention through Projected Matrix Links

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Abstract

Attention selects sources and transmits information through learned interfaces. We formulate these operations for full Gaussian states expressed in learned local coordinate frames. Query and key maps compare lower-dimensional Gaussian observables, and an entropy-regularized categorical objective determines the source weights. At every positive temperature these weights are also the exact posterior of a source label in a normalized model that records query-key compatibility as a binary event. The Gaussian comparison cost is exactly a dot product between query expectation parameters and key natural parameters, offset by a key log-partition bias. When projected key covariances are shared, routing equals whitened dot-product routing with a key-norm penalty, that is, L2 attention in whitened coordinates. The penalty vanishes only for constant key norms or under a compensating state-dependent prior, and neither LayerNorm nor high-dimensional concentration supplies constant norms at the conventional temperature.

Independent sending and receiving maps, dressed by the learned frames, compose into matrix-valued communication links whose endpoint covariance holds for any choice of the maps. A routed link has rank at most $\min(K, \sum_h s_h)$, so rank-deficient heads can sum to an invertible link; under stated rank conditions a head composes without defect exactly when $T_h O_h = I$; and a link of any rank defines a normalized Gaussian channel. The content maps are the value and output maps of standard attention written in local frames, and content aggregation with its residual is the exact predictive mean of a normalized mixture-of-channels kernel. A four-layer model trained on WikiText-103 with a GPT-2 byte-pair vocabulary records token-level test perplexity 31.04 at context 128 in a single-seed feasibility run. [**Data pending:** report the final-protocol result with matched baselines and seed variability (Table 5).]

Keywords: gauge covariance, Gaussian attention, variational routing, matrix links, transformers

1 Introduction

Attention separates source selection from content transmission. Query and key representations determine which inputs are consulted, while value and output maps determine what is communicated (Vaswani et al., 2017). When a representation also carries uncertainty and a local coordinate frame, these operations require compatible transformations of both its mean and covariance. This paper develops a mathematical account of such communication using full Gaussian states, learned comparison interfaces, and matrix-valued links between token positions.

The starting point is selective use of a framed state. A sending map exposes a lower-dimensional observable, and a receiving map places that observable in the recipient's state

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space. Distinct query and key maps define the measurements used to choose sources. Their composition with local frames gives the content link $L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}$, directed from source j to receiver i. Its factors specify accessible directions, gains, and correspondences; they need not be orthogonal projections or group elements. The flat identification $\Omega_{ij} = U_i U_j^{-1}$ appears as a special case. It does not exhaust the geometry of learned communication.

Full Gaussian query and key measurements provide a comparison cost. Minimizing its categorical expectation together with a relative-entropy penalty yields Gibbs routing exactly (Niculae and Blondel, 2017). At every positive temperature the same weights are the exact conditional posterior of a source label in a normalized joint that records query-key compatibility as a binary event, the device used for optimality variables in control as inference (Levine, 2018); at unit temperature they are also a coordinate minimization of a normalized mixture KL. The resulting weights determine a predictive mean through the content links; they do not determine those links, a residual connection, a nonlinear layer, or a vocabulary decoder. We therefore formulate the attention operator directly and state its relation to broader variational inference separately (Blei et al., 2017).

Several parts of this construction have established precedents. Kernel accounts and modern Hopfield networks describe other mathematical structures underlying attention (Tsai et al., 2019; Katharopoulos et al., 2020; Ramsauer et al., 2021). Structured attention networks and latent alignment integrate probabilistic inference into attention (Kim et al., 2017; Deng et al., 2018), variational attention treats the attention vector as a latent variable (Bahuleyan et al., 2018), and attention has been interpreted as structural inference and marginalization (Singh and Buckley, 2023a,b). Gaussian-mixture accounts derive softmax routing as mixture responsibilities, including the key-norm term and a prior that cancels it (Movellan and Gabbur, 2020; Gabbur et al., 2021; Nguyen et al., 2022), and L2 self-attention retains the same key-norm term (Kim et al., 2021). Pro-Attention treats the components of each token vector as samples of a token-specific distribution and scores token pairs by the KL divergence between the estimated distributions (Bae et al., 2022), and Gaussian word embeddings are compared by KL divergence in Vilnis and McCallum (2015). Gaussian-KL routing itself is thus not a priority claim here. Our Gaussian observables are multivariate, retain their full reduced covariance, and enter as laws rather than points, so the query covariance contributes a pair-dependent term that point-query mixtures lack.

Geometric deep learning supplies the representation-theoretic setting (Cohen et al., 2019; Bronstein et al., 2021). SE(3)-Transformers combine attention with features organized by irreducible representations, and Gauge Equivariant Transformer constructs gauge-equivariant multi-head attention on meshes (Fuchs et al., 2020; He et al., 2021). In those constructions the frame is geometric data defined only up to the structure group, so shared maps must be intertwiners. Here the frames are learned for each token and position; they act as a local dressing field (Fournel et al., 2014; François, 2019), and the covariance they carry constrains none of the learned maps (Section 3). Transport of distribution parameters also appears in neuronal gauge accounts (Sengupta et al., 2016; Sengupta and Friston, 2017). These precedents make it necessary to distinguish a frame representation from a communication head. For a product of general linear groups, coordinate blocks are irreducible defining representations, whereas learned heads may access overlapping combinations of those blocks.

The content links obey the endpoint law of lattice link variables, although they are composites of site data rather than independent gauge fields. Conventional lattice connections use group-valued edge maps (Wilson, 1974); linear lattice gauge theory admits more general matrix links with a compact gauge group (Wetterich, 2014), whereas the covariance group here is $\mathrm{GL}(K)$. The attention links form a factorized family of such covariant matrices. Their composition defect, possible rank loss, and routed loop products are distinct from the holonomy of the flat reference identification. Because they factor through an edge space, they are closer to the restriction maps of cellular sheaves and sheaf neural networks (Hansen and Ghrist, 2019; Bodnar et al., 2022). The algebra supplies neither an autonomous lattice action nor a continuum limit.

The contribution is the explicit organization of Gaussian comparison, categorical routing, and predictive content within this matrix-link geometry, together with exact statements about it. Section 2 states the framed Gaussian states, the comparison and content interfaces, and the attention operator in one place. Section 3 proves that the operator factors through frame-invariant reference moments (Theorem 2) and derives the normal form of the evaluated predictor. Section 4 gives the dual-coordinate form of Gaussian-KL routing (Theorem 4) and its reductions to whitened L2 and biased dot-product attention. Section 5 establishes the exact probabilistic status of routing at every temperature and of content aggregation as a predictive mean (Theorem 8). Section 6 develops the rank and composition algebra of the links (Proposition 10) and their relation to lattice gauge theory and cellular sheaves. Section 7 reports a language-model realization and its evaluation limits. The appendices give the smooth-bundle construction, the frame-block representation theory, the positional priors, the prior-bank readout, the relation to a normalized latent-source model, the Gaussian-closure and causal-prediction results, and the experimental record. A separate companion, *Variational Belief Gradients for Gauge-Covariant Gaussian Attention*, develops optional belief-revision objectives and dynamics.

Scope. The results concern the attention operator for full Gaussian states. The content maps, the unit residual coefficient, the normalization, the multilayer perceptron (MLP), and the decoder are taken from the transformer rather than derived. The stored covariance is a static routing state, not propagated uncertainty. No common variational free energy is claimed for the whole predictor, and the construction does not invoke a general Free Energy Principle. The language-model result is a single-seed feasibility record without a matched baseline.

2 Framed Gaussian States and the Attention Operator

This section states the objects and the operator in one place; Sections 3 to 6 prove their properties. A token in a language model is one finite-dimensional realization of an agent; attention-based communication among learned agents has a precedent in targeted multi-agent communication (Das et al., 2019). Appendix A gives the smooth-bundle construction behind the frames, with its geometric picture in Figure 5.

2.1 Framed Gaussian states

Each token position $i \in \{0, \dots, N-1\}$ is an agent whose Gaussian state is expressed in local coordinates on a vector space E_i, identified with the reference space $\mathsf{V} = \mathbb{R}^K$ by an invertible frame $U_i : \mathsf{V} \to E_i$. In the bundle construction of Appendix A, the local section u_i of agent i and one reference trivialization σ_0 are related by $u_i = \sigma_0 \cdot U_i^{-1}$, and the token bundle is pulled back along a constant map, so no connection, curvature, or topology of the contextual base enters the token formulas. The state and its local prior, expressed in the same local coordinates, are

$$q_i = \mathcal{N}(\mu_i, \Sigma_i), \qquad \Sigma_i \succ 0, \tag{1}$$

$$p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i}), \qquad \Sigma_{p,i} \succ 0. \tag{2}$$

Their reference-coordinate representation, in σ_0-coordinates, is obtained through the frame:

$$\widetilde{q}_i = (U_i^{-1})_* q_i = \mathcal{N}(\xi_i, C_i), \qquad \xi_i = U_i^{-1}\mu_i, \quad C_i = U_i^{-1}\Sigma_i U_i^{-\top}. \tag{3}$$

The covariance is an unrestricted positive-definite matrix. The frame determines this coordinate identification; the additional communication interfaces determine which directions another agent receives.

A rechoice $u_i \mapsto u_i \cdot g_i^{-1}$ of the section of agent i, with σ_0 held fixed, changes the coordinates as in Eq. (4) below, by the quotient convention of Eq. (75). These rechoices are independent passive coordinate changes at the token sites. A change of σ_0 is the common reference-basis change $U_i \mapsto U_i a$ of Section 2.2. Under such a passive site change, the same construction is represented by

$$(\mu_i, \Sigma_i, U_i) \mapsto (g_i\mu_i, g_i\Sigma_i g_i^{\top}, g_i U_i), \qquad g_i \in \mathrm{GL}(K). \tag{4}$$

The reference moments (ξ_i, C_i) are unchanged. Reference-space interface parameters are held fixed under this passive action; covectors transform contragrediently. The link endpoint law follows by composing the sending and receiving maps in Section 2.3.

A useful frame parametrization is $U_i = \exp(X_{x_i})\exp(Y_i)$, combining token and positional factors. Its determinants are positive. The passive law above is defined for arbitrary invertible frames; a single real exponential is not a global chart of $\mathrm{GL}^+(K)$, and a product of real exponentials need not have a real logarithm (Culver, 1966). These chart restrictions do not alter the link's endpoint law.

2.2 Communication layers, heads, and comparison spaces

A layer is one communication update applied to the collection of token states. The index ℓ counts successive updates: $q_i^{(\ell)} = \mathcal{N}(\mu_i^{(\ell)}, \Sigma_i^{(\ell)})$ is the state entering layer ℓ, and $q_i^{(\ell+1)}$ is its output. The reference moments $\xi_i^{(\ell)}, C_i^{(\ell)}$ are defined by Eq. (3) for that input state.

Each layer contains H communication heads, indexed by $h = 1, \dots, H$. A head is one channel with its own source comparison and content maps; its weighted messages are combined with those of the other heads. This definition does not require the heads to partition the state or coincide with irreducible frame blocks. For each head choose two real vector spaces,

$$H_h^{\mathrm{cmp}} = \mathbb{R}^{r_h}, \qquad H_h^{\mathrm{val}} = \mathbb{R}^{s_h}. \tag{5}$$

The superscript cmp denotes the comparison space, in which source compatibility is measured; val denotes the content or value space, through which the transmitted message passes. The positive integers r_h and s_h are their dimensions. These spaces are shared across token positions and need not have the same dimension.

The layer's interfaces have the declared types

$$S_{Q,h}^{(\ell)}, S_{K,h}^{(\ell)} : \mathsf{V} \longrightarrow H_h^{\mathrm{cmp}}, \tag{6}$$

$$T_h^{(\ell)} : \mathsf{V} \longrightarrow H_h^{\mathrm{val}}, \qquad O_h^{(\ell)} : H_h^{\mathrm{val}} \longrightarrow \mathsf{V}. \tag{7}$$

The query map S_Q measures the receiver and the key map S_K measures a candidate sender; both have full row rank, hence $r_h \le K$. The value map T extracts content, and the output map O places it in reference state coordinates. Their composition with local frames is developed in Section 2.3. Superscripts (ℓ, h) identify a layer and head, and layer indices are suppressed when one layer is fixed. Matrices act on column vectors, and ij denotes a map from sender j to receiver i.

The maps are specified in the chosen reference basis. Under a common reference-basis change $U_i \mapsto U_i a$, their representations change as $S_Q \mapsto S_Q a$, $S_K \mapsto S_K a$, $T \mapsto Ta$, and $O \mapsto a^{-1}O$, while $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1}C_i a^{-\top}$. Thus the measured laws and the content link are unchanged. Holding the interface matrices fixed while making this reference-basis change would specify a different communication rule.

The query and key Gaussian pushforwards therefore occupy the same comparison space:

$$q_{Q,i}^{(\ell,h)} = \mathcal{N}\big(S_{Q,h}^{(\ell)}\xi_i^{(\ell)}, S_{Q,h}^{(\ell)}C_i^{(\ell)}S_{Q,h}^{(\ell)\top}\big), \tag{8}$$

$$p_{K,j}^{(\ell,h)} = \mathcal{N}\big(S_{K,h}^{(\ell)}\xi_j^{(\ell)}, S_{K,h}^{(\ell)}C_j^{(\ell)}S_{K,h}^{(\ell)\top}\big), \tag{9}$$

$$D_{ij}^{(\ell,h)} = D_{\mathrm{KL}}\big(q_{Q,i}^{(\ell,h)} \| p_{K,j}^{(\ell,h)}\big). \tag{10}$$

Because $C_i^{(\ell)} \succ 0$ and each comparison map has full row rank, the reduced covariance is positive definite: for $y \neq 0$, $y^\top SCS^\top y = (S^\top y)^\top C(S^\top y) > 0$. These are full reduced covariances, including their off-diagonal entries. Thus the construction uses nonsingular distributions in the measured space, rather than treating a singular ambient projection as a K-dimensional Gaussian density.

When $S_Q = S_K = S$, the score compares a common measurement of the two reference beliefs. When $S_Q \neq S_K$, it compares distinct role observables identified in one comparison space. The latter is an additional modeling choice. The maps are learned interface parameters and are generally rectangular; they are not individually invertible gauge transformations. Full input scope means that every head's maps read all K coordinates, as learned projections do in conventional multi-head attention (Vaswani et al., 2017). It does not mean that the comparison dimension equals K.

2.3 The attention operator

One layer combines invariant source weights with covariant content links: each recipient receives a weighted message from its available sources and adds that message to its mean. Source selection is solved at fixed input states; the mean update is an explicit predictive map.

At fixed input beliefs and role maps, source selection minimizes the headwise objective

$$\mathcal{J}_i^{(\ell,h)}(\beta) = \sum_j \beta_j D_{ij}^{(\ell,h)} + \tau_h \sum_j \beta_j \log \frac{\beta_j}{\pi_{ij}^{(h)}}, \qquad \sum_j \beta_j = 1, \tag{11}$$

on the support of $\pi_i^{(h)}$. Differentiating with a normalization multiplier gives

$$\beta_{ij}^{(\ell,h)} = \mathrm{softmax}_j \left(-D_{ij}^{(\ell,h)}/\tau_h + \log \pi_{ij}^{(h)} \right). \tag{12}$$

This is the unique categorical minimizer for finite costs, nonempty allowed support, and positive temperature. The source prior determines the available graph and any positional preference. The temperature is an independent positive scalar; its relation to conventional attention scaling is considered in Section 4.2. Section 5 derives the optimizer and its probabilistic status, and Appendix C gives the causal, window, ALiBi, and relative-position priors.

Content uses independent maps $T_h^{(\ell)} \in \mathbb{R}^{s_h \times K}$ and $O_h^{(\ell)} \in \mathbb{R}^{K \times s_h}$. The former extracts transmitted content; the latter places the head's aggregate in the receiver's full state space. The predictive readout and explicit residual are

$$v_j^{(\ell,h)} = T_h^{(\ell)} \xi_j^{(\ell)}, \tag{13}$$

$$z_i^{(\ell,h)} = \sum_j \beta_{ij}^{(\ell,h)} v_j^{(\ell,h)}, \tag{14}$$

$$m_i^{(\ell)} = U_i \sum_{h=1}^{H} O_h^{(\ell)} z_i^{(\ell,h)}, \tag{15}$$

$$\mu_i^{\mathrm{att},(\ell)} = \mu_i^{(\ell)} + m_i^{(\ell)}. \tag{16}$$

The sum combines all head contributions in the receiving state. The last line defines an attention residual using the pre-update state. Its coefficient is an architectural choice, independent of a belief-gradient step size. As a differentiable operator, the layer depends on source states, routing weights, frame factors, and content maps.

Fix a layer and suppress its index. The content interfaces T_h, O_h of Eq. (7) compose with the local frames to determine a directed map from the sender's state space to the receiver's. Write $A_{Q,i,h} = S_{Q,h} U_i^{-1}$ and $A_{K,i,h} = S_{K,h} U_i^{-1}$ for the local comparison measurements, $\mathsf{S}_{j,h} = T_h U_j^{-1}$ and $\mathsf{R}_{i,h} = U_i O_h$ for the content endpoints, and $G_i = U_i^{-\top} U_i^{-1}$ for the local metric induced by the Euclidean reference inner product. Table 1 collects the geometric data and conventions, and Figure 1 shows the comparison and content interfaces of one head.

The content factors define an ambient matrix link without requiring either factor to be square:

$$L_{ij}^{(h)} = \mathsf{R}_{i,h} \mathsf{S}_{j,h} = U_i O_h T_h U_j^{-1} : E_j \longrightarrow E_i. \tag{17}$$

Under $U_i' = g_i U_i$ and $(\mu_i', \Sigma_i') = (g_i \mu_i, g_i \Sigma_i g_i^\top)$, direct substitution gives

$$\mathsf{S}_{j,h}' = \mathsf{S}_{j,h} g_j^{-1}, \qquad \mathsf{R}_{i,h}' = g_i \mathsf{R}_{i,h}, \qquad L_{ij}^{(h)\prime} = g_i L_{ij}^{(h)} g_j^{-1}. \tag{18}$$

Object	Meaning, domain, and role
N, K, H, B	Number of positions, state dimension, communication heads, and frame blocks
q_i, p_i	Full Gaussian state and local prior on E_i
ξ_i, C_i	Reference moments $U_i^{-1}\mu_i$ and $U_i^{-1}\Sigma_i U_i^{-\top}$
$\pi_{ij}, \beta_{ij}, \tau_h$	Source prior, routing probability, and positive headwise temperature
U_i, $G_i = U_i^{-\top}U_i^{-1}$	Frame $\mathsf{V} \to E_i$ and local metric induced by the Euclidean reference metric
$S_{Q,h}, S_{K,h}$	Full-row-rank maps $\mathsf{V} \to H_h^{\mathrm{cmp}} = \mathbb{R}^{r_h}$ defining comparison observables
T_h, O_h	Sending map $\mathsf{V} \to H_h^{\mathrm{val}} = \mathbb{R}^{s_h}$ and receiving map $H_h^{\mathrm{val}} \to \mathsf{V}$
$A_{Q,i,h}, A_{K,i,h}$	Local measurements $S_{Q,h}U_i^{-1}, S_{K,h}U_i^{-1} : E_i \to H_h^{\mathrm{cmp}}$
$\mathsf{S}_{j,h}, \mathsf{R}_{i,h}$	Content endpoints $T_h U_j^{-1} : E_j \to H_h^{\mathrm{val}}$ and $U_i O_h : H_h^{\mathrm{val}} \to E_i$
$L_{ij}^{(h)}$, $\mathcal{M}_{ij}$	Content link $\mathsf{R}_{i,h}\mathsf{S}_{j,h}$ and routed sum in $\mathrm{Hom}(E_j, E_i)$
Ω_{ij}, Θ_{ij}	Flat reference identification $U_i U_j^{-1}$ and, only when separately specified, invertible connection transport

Table 1: Types and conventions for projection-induced links. Comparison and content spaces are distinct even when their dimensions agree. Layer indices are suppressed.

This is an endpoint covariance law for the complete geometric data. It does not say that a fixed rectangular matrix commutes with every g_i.

The receiving image is $U_i \operatorname{im} O_h$; the invisible sending directions are $\ker(T_h U_j^{-1})$. With the metric G_j in Table 1, the orthogonal complement of that kernel is $U_j \operatorname{im} T_h^\top$. Indeed, $\langle U_j x, U_j y\rangle_{G_j} = x^\top y$, reducing this statement to the Euclidean identity $(\ker T_h)^\perp = \operatorname{im} T_h^\top$. The Fisher metric of the Gaussian location family, $\Sigma_j^{-1} = U_j^{-\top} C_j^{-1} U_j^{-1}$, transforms in the same way without a reference inner product; in that metric the complement is $U_j C_j \operatorname{im} T_h^\top$, and the Gaussian lift of Eq. (64) is the minimal-norm right inverse of the query map. Sending, receiving, and comparison subspaces may overlap. Their dimensions do not impose a partition of the frame, and $O_h T_h$ is generally neither orthogonal nor idempotent.

For invariant scalar routing weights, define

$$\mathcal{M}_{ij} = \sum_h \beta_{ij}^{(h)} L_{ij}^{(h)} = U_i \left(\sum_h \beta_{ij}^{(h)} O_h T_h \right) U_j^{-1}, \tag{19}$$

$$m_i = \sum_j \mathcal{M}_{ij} \mu_j. \tag{20}$$

Equation (18) gives $\mathcal{M}'_{ij} = g_i \mathcal{M}_{ij} g_j^{-1}$ and $m'_i = g_i m_i$. The projected Gaussian routing of Eq. (12) supplies the required scalar invariance: $\xi'_i = \xi_i$ and $C'_i = C_i$, so all comparison laws and their divergences are unchanged. The attention residual $\mu_i + m_i$ consequently has the same vector covariance.

2.4 Depth, token priors, and prediction

At depth ℓ, choose comparison and content maps $S_Q^{(\ell)}, S_K^{(\ell)}, T^{(\ell)}, O^{(\ell)}$. The frame field may be shared across depths while the communication maps vary. An attention residual can be

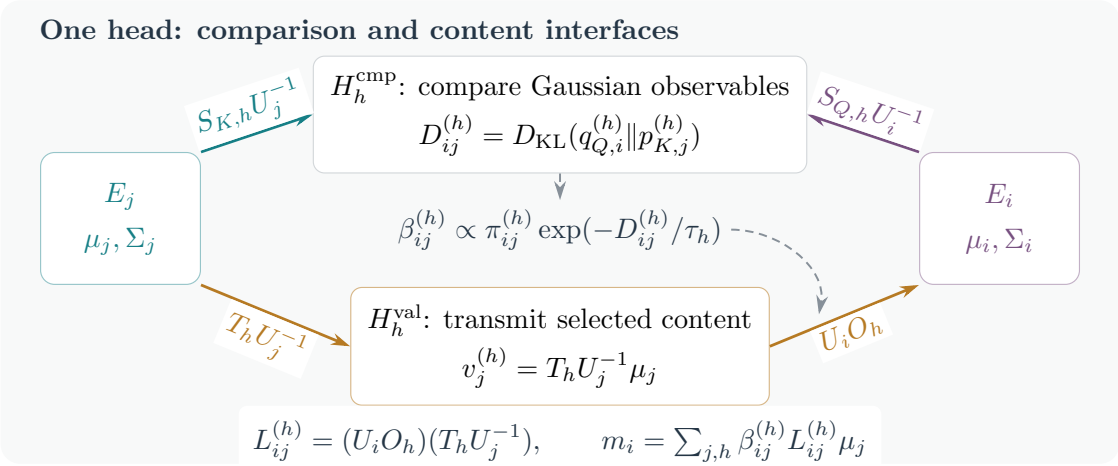


Figure 1: **Separate comparison and content interfaces.** Query and key maps send Gaussian observables into a common comparison space, where their divergence determines the routing weight. The content maps $T_hU_j^{-1}$ and U_iO_h extract and place the transmitted mean. Their composition is the matrix link $L_{ij}^{(h)}$; dashed arrows show scalar routing control, not an additional state map. The Gibbs weights are derived in Section 5.1.

followed by an affine normalization and a coordinate residual MLP. One such composition is

$$u_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}\big(\mu_i^{\mathrm{att},(\ell)}\big), \tag{21}$$

$$\mu_i^{(\ell+1)} = u_i^{(\ell)} + W_2^{(\ell)}\,\mathrm{GELU}\big(W_1^{(\ell)}u_i^{(\ell)} + b_1^{(\ell)}\big) + b_2^{(\ell)}. \tag{22}$$

Here the parameters γ, b of the layer normalization (Ba et al., 2016) are shared across depths, the MLP parameters may differ, and GELU is the Gaussian error linear unit (Hendrycks and Gimpel, 2016). Keeping $\Sigma_i^{(\ell+1)} = \Sigma_i^{(\ell)}$ defines covariance as a routing state rather than the propagated uncertainty of the nonlinear mean map. It remains an input to subsequent comparisons. This construction specifies a predictive architecture; it does not identify the normalized mean with a draw from a transformed Gaussian law.

A coordinatewise activation and fixed affine normalization do not in general commute with arbitrary $\mathrm{GL}(K)$ transformations. Consequently, the covariance of Theorem 2 does not extend automatically to this whole block. Corollary 3 gives the normal form of the evaluated block and its covariant alternatives, and Appendix F.2 makes the Gaussian-closure restriction explicit.

In standard transformers, token embeddings map discrete tokens to continuous vectors via a learned embedding matrix $W_{\mathrm{embed}} \in \mathbb{R}^{V\times K}$. In the present construction, each token type $c \in \{1,\dots,V\}$ is associated with a learned prior distribution:

$$p_c = \mathcal{N}(\mu_p^c, \Sigma_p^c), \qquad \phi_c \in \mathfrak{gl}(K), \tag{23}$$

comprising a prior mean μ_p^c, prior covariance Σ_p^c, and gauge frame coefficients ϕ_c. At the start of each sequence, agent i's belief is initialized to the prior of its token type: $q_i(0) = p_{\text{token}(i)}$. The collection $\{(\mu_p^c, \Sigma_p^c, \phi_c)\}_{c=1}^V$ constitutes a prior bank: a Gaussian embedding table in the sense of Vilnis and McCallum (2015), extended by frame coefficients. If only the means are retained, their table $W_{\text{embed}} = [\mu_p^1, \dots, \mu_p^V]^\top$ has the role of an ordinary embedding matrix. Isotropy alone does not remove the covariance or frame parameters.

The bank initializes the input state independently of the choice of output decoder. Positional frame factors can be composed with its token frame before communication. Using Gaussian templates to encode a token therefore does not require the optional Gaussian-bank vocabulary readout of Appendix D.

For input positions $x_0, \dots, x_{N-1}$, causal source support is $S_i = \{0, \dots, i\}$. The sequence model has the normalized factorization

$$p_\theta(x_0, \dots, x_N) = p_\theta(x_0) \prod_{i=0}^{N-1} p_\theta(x_{i+1} \mid x_{\le i}). \tag{24}$$

A proper initial-token law completes the joint. A linear categorical decoder specifies the next-token factors from the final mean,

$$p_\theta(x_{i+1} = v \mid x_{\le i}) = \text{softmax}_v \left(W_{\text{out}} \mu_i^{(L)} + b_{\text{out}} + \kappa_{\text{uni}} \log p_{\text{uni}} \right), \tag{25}$$

where p_{uni} is a positive vocabulary prior. This decoder is itself a Gibbs readout of shared-covariance templates (Section 5.3), and the token-bank decoder of Appendix D is another choice. Minimizing next-token cross-entropy trains this composed conditional predictor through all of its differentiable components. The observed next token supplies the training target; it is not an input to the causal state construction. Algorithm 1 gives the evaluated realization in the experimental methods. In that evaluation each conditional is computed from a window of at most 128 tokens with stride 32; replacing $x_{\le i}$ by that window leaves every factor normalized, so the reported perplexity is the log-loss of the resulting sequence law.

3 Covariance and the Reference-Coordinate Normal Form

The operator of Section 2.3 is written with frames, but every routing cost and message it computes factors through the reference moments. This section makes that statement exact, derives its consequences, and states what it does and does not constrain.

Lemma 1 (Invariance of divergences under invertible maps) *For probability laws P and Q on $\mathbb{R}^K$ with densities and any $A \in \text{GL}(K)$,*

$$D_{\text{KL}}(A_* P \| A_* Q) = D_{\text{KL}}(P \| Q). \tag{26}$$

Proof Both densities acquire the same Jacobian factor, which cancels in their ratio (Amari, 2016). The same argument applies to f-divergences, and the data-processing inequality applied to A and A^{-1} gives equality for every Rényi order (van Erven and Harremøes, 2014, Thm. 9). ■

This common invertible change of coordinates is distinct from a rank-reducing measurement or a change to the learned interfaces.

Theorem 2 (Reference-moment factorization) *Fix a layer, its interface maps $S_{Q,h}$, $S_{K,h}$, T_h, O_h, its source priors, and its temperatures, and write $M_h = O_h T_h$.*

1. *Every routing cost $D_{ij}^{(h)}$ and weight $\beta_{ij}^{(h)}$ depends on the states only through the reference moments (ξ_j, C_j), and $U_i^{-1}\mu_i^{\mathrm{att}} = \xi_i + \sum_{j,h} \beta_{ij}^{(h)} M_h \xi_j$.*

2. *Under the passive site action of Eq. (4), the routing weights are unchanged, and the routed contributions $\mathcal{M}_{ij}^{(h)} = \beta_{ij}^{(h)} L_{ij}^{(h)}$ and messages transform as*

$$\mathcal{M}_{ij}^{(h)\prime} = g_i \mathcal{M}_{ij}^{(h)} g_j^{-1}, \qquad m_i' = g_i m_i. \tag{27}$$

3. *The operator is unchanged by the common reference change $(U_i, S_{Q,h}, S_{K,h}, T_h, O_h) \mapsto (U_i a, S_{Q,h} a, S_{K,h} a, T_h a, a^{-1} O_h)$ with $a \in \mathrm{GL}(K)$, by $(S_{Q,h}, S_{K,h}) \mapsto (A S_{Q,h}, A S_{K,h})$ with $A \in \mathrm{GL}(r_h)$, and by $(T_h, O_h) \mapsto (B T_h, O_h B^{-1})$ with $B \in \mathrm{GL}(s_h)$. The common reference change also leaves every operation in local coordinates unchanged.*

4. *For a closed walk $\gamma = (i_0, \dots, i_r = i_0)$, the routed loop product $\mathcal{H}_\gamma = \mathcal{M}_{i_0 i_1} \cdots \mathcal{M}_{i_{r-1} i_0}$ satisfies $\mathcal{H}'_\gamma = g_{i_0} \mathcal{H}_\gamma g_{i_0}^{-1}$, and its trace is a polynomial in the routing weights with loop-independent coefficients $\mathrm{Tr}(M_{h_1} \cdots M_{h_r})$. On the causal support $S_i = \{0, \dots, i\}$, every closed walk is a self-loop.*

Proof (1) The comparison laws of Eqs. (8) and (9) are built from $S_{Q,h}\xi_i$, $S_{Q,h} C_i S_{Q,h}^\top$, and the corresponding key moments, so the costs and the Gibbs weights of Eq. (12) depend on the states only through the reference moments. Applying U_i^{-1} to Eq. (16) gives the stated reference form. (2) Substituting the transformed variables into Eq. (3) gives $\xi_i' = \xi_i$ and $C_i' = C_i$, so by (1) the weights are unchanged. The content links obey Eq. (18), which leaves only the external factors g_i and g_j^{-1}, and the message transforms by g_i. (3) Under the common reference change, $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1} C_i a^{-\top}$, so every projected mean $S_{Q,h}\xi_i$, every projected covariance $S_{Q,h} C_i S_{Q,h}^\top$, and the corresponding key moments are unchanged, and $U_i O_h T_h \xi_j$ is unchanged as well. The local coordinates $\mu_i = U_i \xi_i$ are unchanged, so the operations applied in local coordinates are unaffected. Under A, both comparison laws are pushed forward by one invertible map, so Lemma 1 applied in H_h^{cmp} leaves every $D_{ij}^{(h)}$ unchanged. The messages depend on (T_h, O_h) only through M_h. (4) Intermediate factors $g_{i_k}^{-1} g_{i_k}$ cancel in the loop product, and cyclicity of the trace makes its trace invariant. Expanding $U_{i_0}^{-1} \mathcal{H}_\gamma U_{i_0} = M_{i_0 i_1} \cdots M_{i_{r-1} i_0}$ with $M_{ij} = \sum_h \beta_{ij}^{(h)} M_h$ gives the polynomial. Each factor $\mathcal{M}_{i_k i_{k+1}}$ requires $i_{k+1} \le i_k$ on the causal support, so the indices are nonincreasing along the walk and a closed walk is constant. ∎

Tied comparison maps enter only through their row space. The cogredient redundancy of item (3) differs from the contragredient redundancy $(W_Q, W_K) \mapsto (W_Q A, W_K A^{-\top})$ of dot-product scores; the two share only $A \in \mathrm{O}(r_h)$, the residual freedom of whitening in Proposition 5. In particular, a common rescaling of $S_{Q,h}$ and $S_{K,h}$ does not change the routing temperature. Only H_h^{cmp} and H_h^{val} as abstract vector spaces are determined by the model; their bases are internal parameter redundancies. Gradient flow on a loss that depends

on these maps only through the costs and messages therefore conserves $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ and $T_hT_h^\top - O_h^\top O_h$ (Du et al., 2018; Kunin et al., 2021), whereas adaptive optimizers without weight decay can drift along these directions.

What the covariance does and does not constrain. Theorem 2 uses only the fact that routing and content depend on the states through the reference moments (ξ_i, C_i). Choosing $g_i = U_i^{-1}$ therefore removes the frames, and the operator becomes Eq. (40) with every $U_i = I$. Any operator with this property is covariant in the same sense, including dot-product attention on dressed means and any coordinate map applied as $U_i G(U_i^{-1}\cdot)$. In the terminology of the dressing-field method (Fournel et al., 2014), U_i is a local dressing field, and the local $\mathrm{GL}(K)$ covariance of the attention operator is artificial in the sense of François (2019, Sec. 2.2). It is the kind of covariance that, as Kretschmann objected in the case of general relativity, a formulation can acquire by introducing auxiliary fields (Norton, 1993). Equivalently, the learned frames fix a global trivialization of the token bundle, an $\{e\}$-structure in the sense of Weiler et al. (2021, Sec. 2.1.1), and the covariance places no constraint on S_Q, S_K, T, or O. Gauge-equivariant convolution and attention on manifolds instead treat the frame as geometric data defined only up to the structure group, so their shared maps must be intertwiners (Cohen et al., 2019, Sec. 2.4; He et al., 2021, Secs. 4.3 and 4.4). For $G = \prod_b \mathrm{GL}(d_b)$ that requirement would confine each O_hT_h to $\mathrm{End}_G(\mathsf{V}) = \bigoplus_b \mathbb{R}I_{d_b}$, so every content link would reduce to $\sum_b c_b \Omega_{ij}^{(b)}$, and full-row-rank comparison maps would reduce routing to blockwise frame-only divergences. The low rank, block mixing, and composition defect of the links studied in Section 6 are available because the frames are learned.

The evaluated architecture applies its normalization, MLP, and decoder in local coordinates, and there the learned frames act beyond the attention operator.

Corollary 3 (Normal form of the evaluated predictor) *In reference coordinates the predictor of Section 2.4 reads*

$$\xi_i^{(0)} = U_i^{-1}\mu_{p,x_i}, \qquad C_i = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}, \qquad \xi_i^{(\ell+1)} = U_i^{-1}\Phi^{(\ell)}\big(U_i\xi_i^{\mathrm{att},(\ell)}\big), \tag{28}$$

with $\Phi^{(\ell)} = (\mathrm{id} + \mathrm{MLP}^{(\ell)}) \circ \mathrm{LN}$, with the attention step frame-free given the routing weights, and with decoder logits $W_{\mathrm{out}}U_i\xi_i^{(L)}$ plus the class biases.

Proof Apply U_i^{-1} to Eqs. (22) and (25), use item (1) of Theorem 2 for the attention step, and use covariance passthrough and depth-independent frames for C_i. ∎

The frames therefore act through four channels: a multiplicative token-and-position embedding; a routing covariance that is static per token type and position, because covariance passes through unchanged and the frames are fixed through the stack; a token- and position-specific conjugation of the shared normalization and MLP; and a token- and position-specific decoder $W_{\mathrm{out}}U_i$. Figure 2 depicts these roles. Covariant alternatives to the last two channels exist. Conjugating the coordinate block by the frame, $\mu_i \mapsto U_i G(U_i^{-1}\mu_i)$, or scaling each frame block by a function of the invariant $\mu_b^\top \Sigma_{bb}^{-1}\mu_b$, gives a block that is covariant under the passive site action. With such a block and a decoder acting on ξ_i, the whole predictor depends on the frames only through $\xi_i^{(0)}$ and C_i.

PLACEHOLDER M1: SCHEMATIC TO DRAW

Left panel: one evaluated block in local coordinates, $\mu_i \mapsto \mu_i + m_i \mapsto$ LayerNorm $\mapsto$ MLP $\mapsto$ decoder, with the frame entering at $\xi_i = U_i^{-1}\mu_i$ and at $m_i = U_i \sum_h O_h z_i^{(h)}$.

Right panel: the same computation after the site transformation $g_i = U_i^{-1}$. Attention acts on (ξ_i, C_i) without frames; the frames remain only in the input moments $\xi_i^{(0)} = U_i^{-1}\mu_{p,x_i}$, in the routing covariance $C_i = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}$ fixed by token type and position, in the conjugated block $U_i^{-1}\Phi^{(\ell)}(U_i(\cdot))$ with $\Phi^{(\ell)} = (\mathrm{id} + \mathrm{MLP}^{(\ell)}) \circ \mathrm{LayerNorm}$, and in the decoder $W_{\mathrm{out}}U_i$.

TO DRAW: vector schematic in the notation of Table 1; mark each frame role with the row of Table 4 that removes it.

Figure 2: Reference-coordinate form of the evaluated architecture. Every routing cost and message of the attention operator factors through the reference moments (ξ_i, C_i), so the site transformation $g_i = U_i^{-1}$ removes the frames from that operator exactly, as Eq. (27) implies. In the evaluated model the frames therefore act only through the input moments, a routing covariance fixed by token type and position, the conjugation of the coordinate normalization and MLP, and the decoder.

4 Gaussian-KL Routing and Its Transformer Reductions

The correspondence with standard attention starts from the learned comparison and content maps, rather than replacing them by the flat identification Ω_{ij}. We establish an exact dual-coordinate form of the routing cost (Theorem 4), its reduction under stated covariance and bias conditions, and an exact identity for the resulting value/output aggregation. These are conditional equalities of attention operators; Table 2 summarizes them by status.

4.1 The full Gaussian score and its query–key cross term

Fix a layer, a head, and its nonempty allowed source sets; suppress the layer and head indices. Write $r = r_h$ and define the projected moments

$$a_i = S_Q\xi_i, \quad b_j = S_K\xi_j, \quad V_{Q,i} = S_QC_iS_Q^\top, \quad V_{K,j} = S_KC_jS_K^\top. \tag{29}$$

The state covariances are positive definite and both comparison maps have full row rank, so $V_{Q,i}, V_{K,j} \succ 0$. The Gaussian KL formula gives

$$\begin{aligned} D_{ij} = \frac{1}{2}\big[\,&\mathrm{Tr}(P_jV_{Q,i}) + a_i^\top P_j a_i - 2a_i^\top P_j b_j + b_j^\top P_j b_j - r \\ &+ \log\det V_{K,j} - \log\det V_{Q,i}\big], \qquad P_j = V_{K,j}^{-1}. \end{aligned} \tag{30}$$

Its mean compatibility has the dot-product factorization $a_i^\top(P_jb_j)$; the KL contains the negative of this term. In ambient local coordinates it is $\mu_i^\top B_{ij}\mu_j$ with the comparison tensor of Eq. (62). Thus independent learned S_Q, S_K already supply distinct query and

key observables. Neither is required to be a gauge-group element. The other terms retain covariance information: when P_j varies with the source, $a_i^\top P_j a_i$ and $\mathrm{Tr}(P_j V_{Q,i})$ are generally pair dependent and cannot be discarded as row constants.

All of these terms are organized by one exact identity.

Theorem 4 (Dual-coordinate form of Gaussian-KL routing) *Write the query law in expectation coordinates and the key law in natural coordinates,*

$$\eta_{Q,i} = \big(a_i, V_{Q,i} + a_i a_i^\top\big), \qquad \theta_{K,j} = \big(P_j b_j, -\tfrac{1}{2}P_j\big), \qquad \langle \eta, \theta\rangle = \eta_1^\top \theta_1 + \mathrm{Tr}(\eta_2 \theta_2), \tag{31}$$

and let $A(\theta_{K,j}) = \frac{1}{2} b_j^\top P_j b_j + \frac{1}{2}\log\det V_{K,j} + \frac{r}{2}\log 2\pi$ be the Gaussian log-partition, with conjugate $A^*(\eta_{Q,i}) = -H(q_{Q,i})$. Then

$$D_{ij} = A(\theta_{K,j}) + A^*(\eta_{Q,i}) - \langle \eta_{Q,i}, \theta_{K,j}\rangle, \tag{32}$$

and the categorical optimizer is exactly $\beta_{ij} = \mathrm{softmax}_j\big(\langle \eta_{Q,i}, \theta_{K,j}\rangle/\tau - A(\theta_{K,j})/\tau + \log \pi_{ij}\big)$: a dot product in $\mathbb{R}^{r+r(r+1)/2}$ between query expectation parameters and key natural parameters, with the key log-partition as a source bias. No covariance hypothesis is needed.

Proof With $-H(q_{Q,i}) = -\frac{1}{2}\log\det(2\pi e V_{Q,i})$, expanding the right side of Eq. (32) reproduces Eq. (30) term by term. This is the mixed form of the divergence between members of an exponential family (Wainwright and Jordan, 2008, Sec. 5.2, Eq. (5.11)). The term $A^*(\eta_{Q,i})$ is constant on row i and cancels in the softmax. ■

Proposition 5 is the case in which the second-order key coordinate $-\frac{1}{2}P_j$ is shared, so that only the mean block $a_i^\top V_*^{-1} b_j$ of the pairing varies with the source.

Proposition 5 (Projected Gaussian reduction to biased dot-product routing) *Suppose all keys used by a head have the same projected covariance $V_{K,j} = V_* \succ 0$. Let $\tau > 0$, let each prior row be a probability distribution with $\pi_{ij} > 0$ on its allowed support, and let costs be the uncapped Gaussian divergences. Define $\mathsf{q}_i = V_*^{-1/2} a_i$ and $\mathsf{k}_j = V_*^{-1/2} b_j$, using the symmetric positive-definite inverse square root. Then the exact categorical optimizer is*

$$\beta_{ij} = \mathrm{softmax}_j\left(\frac{\mathsf{q}_i^\top \mathsf{k}_j}{\tau} - \frac{\|\mathsf{k}_j\|^2}{2\tau} + \log \pi_{ij}\right). \tag{33}$$

The query covariances may differ across receivers. If, more strongly, $V_{Q,i} = V_{K,j} = \sigma_h^2 I_r$, then $D_{ij} = \|a_i - b_j\|^2/(2\sigma_h^2)$.

Proof Set $P_j = V_*^{-1}$ in Eq. (30). The trace, query quadratic, dimension, and determinant terms are independent of j, except for $b_j^\top V_*^{-1} b_j$ and the cross term. Thus

$$D_{ij} = c_i + \tfrac{1}{2}\|\mathsf{k}_j\|^2 - \mathsf{q}_i^\top \mathsf{k}_j, \tag{34}$$

for a row constant c_i. Substitution into the categorical solution $\mathrm{softmax}_j(-D_{ij}/\tau + \log \pi_{ij})$ removes c_i. Under the stronger isotropic assumption the trace and log-determinant terms cancel and the remaining quadratic is the stated squared distance. ■

Since $D_{ij} = \frac{1}{2}\|\mathsf{q}_i - \mathsf{k}_j\|^2$ up to a row constant, routing with shared projected key covariance is L2 attention in whitened coordinates; under the stronger isotropic hypothesis it is the L2 self-attention of Kim et al. (2021). Dot-product attention is recovered only when projected key norms are constant or compensated (Section 4.2).

Remark 6 (Relation to Gaussian-mixture attention) *Since* $D_{\mathrm{KL}}(q\|p) = -\mathbb{E}_q \log p - H(q)$ *and* $H(q_{Q,i})$ *is constant on row* i*,*

$$\beta_{ij} = \mathrm{softmax}_j\big(\log \pi_{ij} + \mathbb{E}_{q_{Q,i}}[\log p_{K,j}]/\tau\big), \tag{35}$$

the mean-field responsibility of a Gaussian mixture under an uncertain query. For a point query at a_i *it is the mixture responsibility of Movellan and Gabbur (2020) and Nguyen et al. (2022), and Eq.* (36) *below is the prior with which Movellan and Gabbur (2020, Eqs. 16 to 21) recover dot-product attention. The query covariance adds* $-\mathrm{Tr}(P_j V_{Q,i})/(2\tau)$*, which depends on the pair whenever* P_j *varies with the source.*

The hypotheses of Proposition 5 concern the projected distributions, not merely isotropic ambient states. Even if $C_i = \sigma^2 I_K$, a learned map has covariance $\sigma^2 S_K S_K^\top$, which need not be isotropic. Shared $C_i = C$ across tokens does give a shared key covariance for fixed S_K and therefore satisfies the weaker proposition. It places no equality requirement on S_Q and S_K. Whitening then defines the effective learned maps $\widetilde{S}_Q = V_*^{-1/2} S_Q$ and $\widetilde{S}_K = V_*^{-1/2} S_K$. The simple choice $K = 2r$, $S_Q = [I_r\ 0]$, $S_K = [0\ I_r]$, and $C_i = \sigma^2 I_K$ demonstrates that distinct query and key observables can satisfy even the stronger isotropic assumptions. A trained state with source-dependent projected covariance need not satisfy either specialization.

4.2 Exact bias cancellation, approximation, and temperature

Equation (33) becomes ordinary dot-product attention with the same source prior exactly when the squared key norms are constant on the allowed support. More generally, a desired base prior π^0_{ij} can be reproduced algebraically by choosing

$$\pi_{ij} = \frac{\pi^0_{ij} \exp(\|\mathsf{k}_j\|^2/(2\tau))}{\sum_{k\in S_i} \pi^0_{ik} \exp(\|\mathsf{k}_k\|^2/(2\tau))}. \tag{36}$$

This is an explicit state-dependent compensation. It is not a positional prior fixed independently of the current key states, and it is not assumed to be implemented. A minimal counterexample shows why the condition matters: for $r = 1$, $\tau = 1$, $a_i = 0$, keys $b_1 = 0, b_2 = 1$, shared unit variance, and uniform prior, ordinary dot-product routing gives $(1/2, 1/2)$ while Gaussian routing gives $(1, e^{-1/2})/(1 + e^{-1/2})$.

An approximation can be stated by bounding the remaining bias.

Proposition 7 (Sharp routing perturbation bound) *Under the hypotheses of Proposition 5, for any row constant* c_i *define*

$$\varepsilon_i = \max_{j\in S_i} \frac{|\|\mathsf{k}_j\|^2 - c_i|}{2\tau}, \qquad p_{ij} = \mathrm{softmax}_j\left(\mathsf{q}_i^\top \mathsf{k}_j/\tau + \log \pi_{ij}\right). \tag{37}$$

Then

$$\sum_j |\beta_{ij} - p_{ij}| \le 2\tanh(\varepsilon_i/2) \le \min(\varepsilon_i, 2), \tag{38}$$

with equality for two sources with $p_{i1} = 1/(1 + e^{\varepsilon_i})$ *and key-norm offsets of* $\pm\varepsilon_i$*. The best row constant is the midrange, which gives* $\varepsilon_i = (\max_j \|\mathsf{k}_j\|^2 - \min_j \|\mathsf{k}_j\|^2)/(4\tau)$*.*

Proof By Eq. (33), $\beta_{ij} = p_{ij}e^{\delta_{ij}}/\sum_k p_{ik}e^{\delta_{ik}}$ with $\delta_{ij} = -(\|\mathsf{k}_j\|^2 - c_i)/(2\tau)$, so $|\delta_{ij}| \le \varepsilon_i$. For $A \subseteq S_i$, the derivative of $\beta_i(A)$ in δ_{ik} is positive for $k \in A$ and negative otherwise, so $\beta_i(A) - p_i(A)$ is largest at $\delta_{ij} = \varepsilon_i$ on A and $\delta_{ij} = -\varepsilon_i$ off A, where it equals $x(1-x)(e^{2\varepsilon_i} - 1)/[1 + x(e^{2\varepsilon_i} - 1)]$ with $x = p_i(A)$. The maximum over x is attained at $x = 1/(1 + e^{\varepsilon_i})$ and equals $\tanh(\varepsilon_i/2)$. Since $\sum_j |\beta_{ij} - p_{ij}| = 2\max_A[\beta_i(A) - p_i(A)]$, the bound follows, and $2\tanh(\varepsilon_i/2) \le \min(\varepsilon_i, 2)$ because tanh is bounded by one and by its argument. The midrange minimizes ε_i over c_i. ■

The ratio bound $e^{-2\varepsilon_i} \le \beta_{ij}/p_{ij} \le e^{2\varepsilon_i}$ remains available as a multiplicative statement. Therefore uniform control $\varepsilon_i \to 0$ suffices for rowwise convergence, including when the number of keys varies. For the conventional scale $\tau = \sqrt{r}$, this requires maximum squared-norm dispersion $o(\sqrt{r})$ on the row support. High-dimensional relative concentration alone is insufficient: for independent standard Gaussian vectors, the squared norm has variance $2r$ and the dot product of independent vectors has variance r. Both residuals retain order-one fluctuations after division by $\sqrt{r}$. Coordinate LayerNorm followed by an arbitrary learned key map does not impose the required projected-key norm constancy.

Standard attention uses the comparison dimension r_h in its scale $\sqrt{r_h}$ (Vaswani et al., 2017, Sec. 3.2.1). After whitening, the cross term has this conventional scale when $\tau_h = \sqrt{r_h}$. For arbitrary positive τ_h, rescaling both effective query and key by $(\sqrt{r_h}/\tau_h)^{1/2}$ writes the same cross term with denominator $\sqrt{r_h}$. Covariance scale has already entered through whitening and must not be counted a second time as temperature. Numerical modification of the KL cost lies outside the uncapped score identity unless it leaves every relevant cost unchanged.

Geometric dilation must also be separated from statistical sharpness. A passive rescaling of one frame block (Appendix B) transforms its mean, covariance, and frame together as $(\mu, \Sigma, U) \mapsto (c\mu, c^2\Sigma, cU)$, $c \neq 0$. The reference moments and all routing costs are unchanged. At fixed projected means, changing a shared isotropic comparison covariance $\sigma_h^2 I$ instead changes the statistical model and gives $D_{ij}^{(h)} = \|S_{Q,h}\xi_i - S_{K,h}\xi_j\|^2/(2\sigma_h^2)$. The logits then depend on σ_h^2 and τ_h only through $\sigma_h^2\tau_h$, so the two are not separately identifiable from routing. For general projected covariances, trace and determinant terms prevent reducing every covariance change to one scalar temperature.

4.3 Value, output, and residual links

For any routing weights, independently of the dot-product specialization, predictive content has the exact reference-coordinate form

$$v_j^{(h)} = T_h\xi_j, \qquad z_i^{(h)} = \sum_j \beta_{ij}^{(h)} v_j^{(h)}, \tag{39}$$

$$U_i^{-1}\mu_i^{\mathrm{att}} = \xi_i + \sum_h O_h z_i^{(h)} = \xi_i + \sum_{j,h} \beta_{ij}^{(h)} O_h T_h \xi_j. \tag{40}$$

Let $O = [O_1 \ \cdots \ O_H]$ and concatenate the $z_i^{(h)}$. Then $\sum_h O_h z_i^{(h)} = O\,\mathrm{concat}_h(z_i^{(h)})$, the usual multi-head output projection (Vaswani et al., 2017, Sec. 3.2.2). With column-vector conventions, the dictionary is $W_V^{(h)} = T_h$ and output blocks $W_O^{(h)} = O_h$; the matrices in the original row-vector transformer convention are their transposes. In local coordinates

the same operation is $\sum_{j,h} \beta_{ij}^{(h)} L_{ij}^{(h)} \mu_j$. Thus the content link is the framed value/output composition, not the query–key compatibility tensor.

Each head can read all K coordinates and write into all K output coordinates. The latent channel dimension limits the rank of an individual contribution, not its input or output support. Communication heads need not coincide with a partition into gauge-frame blocks; full-width maps can mix those blocks even when every U_i is block diagonal. Equation (40), together with Proposition 5 and its bias conditions, supplies the conditional attention-layer correspondence. The explicit residual is part of this architecture; Theorem 8 reads it as the persistence term of the predictive kernel of Eq. (49), and Section 7.2 reports that it is numerically inert in the trained model.

4.4 Restricted rotary correspondence and the remaining stack

Rotary position embedding (RoPE) applies position-dependent orthogonal rotations after query and key projection (Su et al., 2024). For an even comparison dimension, let R_i be a block-diagonal matrix of planar rotations with angles proportional to position, so $R_i^\top R_j = R_{j-i}$. A declared comparison-space construction $a_i \mapsto R_i a_i$, $b_j \mapsto R_j b_j$ has cross term

$$(R_i a_i)^\top (R_j b_j) = a_i^\top R_{j-i} b_j = \xi_i^\top S_Q^\top R_{j-i} S_K \xi_j. \tag{41}$$

Rotating both projected laws gives, by Eq. (26), $D_{\mathrm{KL}}\big((R_i)_* q_{Q,i} \| (R_j)_* p_{K,j}\big) = D_{\mathrm{KL}}\big(q_{Q,i} \| (R_{j-i})_* p_{K,j}\big)$, a relative-position score for arbitrary projected covariances. The dot-product form with an unchanged key-norm bias requires only that the shared key covariance commute with every R_δ; with distinct rotation frequencies this means $V_* = \bigoplus_k \lambda_k I_2$ over the rotation planes. Query covariance is unrestricted. If the rotations act on anisotropic projected laws rather than on means alone, the trace and query-quadratic terms of Eq. (30) also become functions of $i - j$ and contribute relative-position biases.

This identity does not automatically identify position rotations inside an ambient frame with ordinary post-projection RoPE. Moving an ambient positional map through S_Q or S_K requires the corresponding intertwining identities, such as $S_Q e^{-Y_i} = R_i S_Q$ and $S_K e^{-Y_i} = R_i S_K$ for the positional factor of $U_i = e^{X_{x_i}} e^{Y_i}$; imposing the identity on the complete U_i^{-1} would require every token factor to act trivially on the comparison row spaces. Arbitrary learned maps need not satisfy them. In addition, ordinary RoPE leaves value aggregation unrotated. The separated comparison and content maps allow this design to be specified, but one shared ambient positional frame acting through every U_i^{-1} does not establish it. The relative rotation is a flat group-valued factor; the full projected compatibility remains a generally low-rank bilinear tensor, and the content link remains a separate operator.

Covariance may be retained as a routing state in a predictive architecture. Passing it unchanged through the mean residual does not compute the covariance of a Gaussian mixture or propagate uncertainty through normalization and the MLP. For example, a declared mixture of content laws with means v_j and covariances W_j would have covariance $\sum_j \beta_j [W_j + (v_j - \bar{v})(v_j - \bar{v})^\top]$, where $\bar{v} = \sum_j \beta_j v_j$. The predictive mean identity does not require this additional propagation rule; Eq. (50) gives its multi-head form for the predictive kernel.

Correspondence	Exact scope or remaining condition
Derived under stated conditions	
Gaussian categorical routing	Exact optimizer for fixed costs, allowed support, and positive temperature; exact label posterior under a binary compatibility record
Query/key dot product	Exact dot product in dual coordinates, Eq. (32); reduces to the r_h-dimensional mean block under shared key covariance
Ordinary attention weights	Shared key covariance, matched scale, and constant or compensated key-norm bias; otherwise L2 attention in whitened coordinates
RoPE score	Relative dependence exact for arbitrary projected covariances when both laws rotate; dot-product form requires a shared key covariance that commutes with the rotations
Identical by construction	
Value/output aggregation	T_h and O_h are the value and output maps; framed content-link identity at any fixed routing
Multi-head organization	Full-width maps and output concatenation identity; no frame-block partition is required
Not derived (architectural choices)	
Predictive residual	Unit-coefficient term, read as the persistence term of a predictive kernel; numerically inert after training in the reported run
Covariance, normalization, MLP, decoder	Separate processing choices; the linear decoder is a Gibbs readout of shared-covariance templates

Table 2: Transformer correspondences for the projection-based architecture, grouped by status. The equalities concern the specified operators, not an identity between unrestricted model families.

5 Probabilistic Status of Routing and Content

The Gibbs row of Eq. (12) and the content update of Eq. (16) were stated as operations. This section derives the row from a declared local source model, shows that it is an exact posterior at every temperature, and shows that the content update with its residual is an exact predictive mean.

5.1 Variational source selection

Fix a layer, head, and receiver, suppressing those indices. Let $q_{Q,i}(y)$ be the projected query law and $p_{K,j}(y)$ the candidate key laws. On a nonempty allowed support S_i, take positive probabilities π_{ij} summing to one. The normalized local source model and factorized recognition law are

$$\mathbb{P}_i(y, J = j) = \pi_{ij} p_{K,j}(y), \tag{42}$$

$$\mathbb{Q}_{i,\beta}(y, J = j) = \beta_{ij} q_{Q,i}(y). \tag{43}$$

Here the Gaussian states are fixed values at the given context: they are computed from the prefix before routing and enter the local joint as data, so $p_{K,j}$ reads a state value rather

than a recognition law being optimized, and β_i is the only recognition variable. The receiver enters only through the query law, clamped as the recognition marginal of y. Hold the query law fixed during categorical minimization. Expanding the density ratio gives

$$D_{\mathrm{KL}}(\mathbb{Q}_{i,\beta}\|\mathbb{P}_i) = \sum_{j\in S_i} \beta_{ij} D_{ij} + \sum_{j\in S_i} \beta_{ij} \log \frac{\beta_{ij}}{\pi_{ij}}, \tag{44}$$

$$D_{ij} = D_{\mathrm{KL}}(q_{Q,i}\|p_{K,j}). \tag{45}$$

The categorical entropy belongs to this variational objective. The direction of D_{ij} is fixed by the declared joint and the exclusive divergence: the key laws are the generative mixture components and the query law is the recognition factor. The inclusive divergence gives $D_{\mathrm{KL}}(\mathbb{P}_i\|\mathbb{Q}_{i,\beta}) = D_{\mathrm{KL}}(\pi_i\|\beta_i) + \sum_j \pi_{ij} D_{\mathrm{KL}}(p_{K,j}\|q_{Q,i})$, whose minimizer is $\beta_i = \pi_i$ for all costs. The reverse cost $D_{\mathrm{KL}}(p_{K,j}\|q_{Q,i})$ follows from the joint $\pi_{ij} q_{Q,i}(y)$ with recognition $\beta_{ij} p_{K,j}(y)$; its reduction to a dot product requires a shared query covariance and retains a source-dependent bias. Symmetric divergences require both projected covariances to be shared.

At a declared temperature $\tau > 0$, use

$$\mathcal{A}_i(\beta) = \sum_{j\in S_i} \beta_{ij} D_{ij} + \tau D_{\mathrm{KL}}(\beta_i\|\pi_i). \tag{46}$$

The Hessian $\tau\,\mathrm{diag}(1/\beta_{ij})$ is positive definite on nonzero simplex tangent directions. The Lagrange condition is $D_{ij} + \tau(\log(\beta_{ij}/\pi_{ij}) + 1) + \lambda_i = 0$. For finite costs the unique minimizer is interior on S_i:

$$\beta^*_{ij} = \frac{\pi_{ij} \exp(-D_{ij}/\tau)}{\sum_{k\in S_i} \pi_{ik} \exp(-D_{ik}/\tau)}, \tag{47}$$

$$\beta^*_{ij} = \mathrm{softmax}_{j\in S_i}(-D_{ij}/\tau) \quad \text{for uniform } \pi_i. \tag{48}$$

Substitution gives $-\tau \log Z_i$, with $Z_i = \sum_{j\in S_i} \pi_{ij} \exp(-D_{ij}/\tau)$, by the usual entropy-regularized variational calculation (Boyd and Vandenberghe, 2004; Wainwright and Jordan, 2008; Niculae and Blondel, 2017). At $\tau = 1$ this is the KL of the declared local joint; general temperature changes that objective. The categorical minimizer needs no gradient-based update of q_i. Local mixture normalization does not supply a population model or a language-model observation likelihood.

Complete-state alignment is a special comparison: for $S_Q = S_K = I_K$, common push-forward by U_i gives $D_{ij} = D_{\mathrm{KL}}(q_i\|(\Omega_{ij})_* q_j)$. Independent lower-dimensional role measurements therefore extend the comparison beyond this frame-only case. For tied maps, $D_{ij} \le D_{\mathrm{KL}}(q_i\|(\Omega_{ij})_* q_j)$ by the data-processing inequality (van Erven and Harremøes, 2014, Thm. 9), so a shared comparison coarse-grains complete-state alignment; for $S_Q \neq S_K$ no such bound holds, and identical states can have $D_{ij} > 0$.

5.2 Exact posterior and predictive mean

Theorem 8 (Exact probabilistic status of routing and content) *Fix a layer and its input states, and let $D^{(h)}_{ij} \ge 0$ be the routing costs.*

1. *For every $\tau_h > 0$, give each receiver and head a source label $J_i^{(h)}$ with prior $\pi_i^{(h)}$ and a binary compatibility record $o_i^{(h)}$ with $P(o_i^{(h)} = 1 \mid J_i^{(h)} = j) = \exp(-D_{ij}^{(h)}/\tau_h)$. The joint is normalized on any source graph, and $P(J_i^{(h)} = j \mid o_i^{(h)} = 1) = \beta_{ij}^{(h)}$ of Eq. (12). The objective of Eq. (11) is τ_h times the label free energy $D_{\mathrm{KL}}(\beta \| \pi_i^{(h)}) - \sum_j \beta_j \log P(o_i^{(h)} = 1 \mid J_i^{(h)} = j)$, whose minimum is the surprisal $-\log P(o_i^{(h)} = 1)$.*

2. *Let the head labels be independent with laws $\beta_i^{(h)}$, and take the directed kernel*

$$k_i' \mid k, J_i \sim \mathcal{N}\Big(k_i + \sum_h L^{(h)}_{iJ_i^{(h)}} k_{J_i^{(h)}}, Q_i\Big), \qquad Q_i \succeq 0, \tag{49}$$

which is normalized for any link rank, as a degenerate Gaussian when Q_i is singular. At state values $k_j = \mu_j$ its predictive mean is μ_i^{att} of Eq. (16). With independent source states $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$, the self edge included, its moment-matched covariance in reference coordinates is

$$\begin{aligned} C_i^{\mathrm{att}} = \widetilde{Q}_i + C_i + \sum_h \beta_{ii}^{(h)} \big(M_h C_i + C_i M_h^\top\big) + \sum_j \sum_{h,h'} c_{ij}^{hh'} M_h C_j M_{h'}^\top \\ + \sum_h \Big[\sum_j \beta_{ij}^{(h)} n_{ij}^{(h)} n_{ij}^{(h)\top} - \bar{n}_i^{(h)} \bar{n}_i^{(h)\top}\Big], \end{aligned} \tag{50}$$

where $\widetilde{Q}_i = U_i^{-1} Q_i U_i^{-\top}$, $M_h = O_h T_h$, $c_{ij}^{hh} = \beta_{ij}^{(h)}$, $c_{ij}^{hh'} = \beta_{ij}^{(h)} \beta_{ij}^{(h')}$ for $h \neq h'$, $n_{ij}^{(h)} = M_h \xi_j$, and $\bar{n}_i^{(h)} = \sum_j \beta_{ij}^{(h)} n_{ij}^{(h)}$. The local covariance $U_i C_i^{\mathrm{att}} U_i^\top$ obeys the passive law.

Proof (1) Because $D_{ij}^{(h)} \geq 0$, each record kernel takes values in $(0, 1]$ and is normalized. Records attached to different receivers and heads are conditionally independent children of the states and labels, so the joint is normalized. Bayes' rule gives $P(J = j \mid o = 1) = \pi_{ij} e^{-D_{ij}/\tau} / \sum_k \pi_{ik} e^{-D_{ik}/\tau}$, which is Eq. (12). Since $-\log P(o = 1 \mid J = j) = D_{ij}/\tau$, the label free energy equals τ^{-1} times the objective, and its minimum is $-\log \sum_j \pi_{ij} e^{-D_{ij}/\tau} = -\log P(o = 1)$. (2) Write $X = k_i + \sum_h L^{(h)}_{iJ^{(h)}} k_{J^{(h)}}$. Conditioning on the labels gives $\mathbb{E}[X \mid J] = \mu_i + \sum_h L^{(h)}_{iJ^{(h)}} \mu_{J^{(h)}}$, whose average over the independent labels is μ_i^{att}. By the law of total covariance, $\operatorname{Cov} X = \mathbb{E} \operatorname{Cov}(X \mid J) + \operatorname{Cov} \mathbb{E}(X \mid J)$. Given the labels, each distinct source s contributes $A_s \Sigma_s A_s^\top$, with $A_i = I + \sum_h \mathbf{1}[J^{(h)} = i] L_{ii}^{(h)}$ and $A_s = \sum_h \mathbf{1}[J^{(h)} = s] L_{is}^{(h)}$ for $s \neq i$. Averaging the indicator products with $\mathbb{E}\big(\mathbf{1}[J^{(h)} = s]\mathbf{1}[J^{(h')} = s]\big) = c_{is}^{hh'}$ gives the terms in C_i, $\beta_{ii}^{(h)}$, and $c_{ij}^{hh'}$, and independence of the labels across heads gives the spread term. Conjugating by U_i^{-1} removes every frame, since $L_{ij}^{(h)} \Sigma_j L_{ij}^{(h')\top} = U_i M_h C_j M_{h'}^\top U_i^\top$, and adding Q_i completes the kernel covariance. ∎

The record is a declared modeling event of the kind used for optimality variables in control as inference (Toussaint and Storkey, 2006; Kappen et al., 2012; Levine, 2018); it is not an observed datum. Exactness of this kind is available for any nonnegative cost, so item (1) shows that Gibbs routing is consistent with exact inference, not that attention must be inference. Covariance passthrough retains only the term C_i of Eq. (50). The derivation of

item (2) leaves T_h, O_h, and Q_i free; it declares the unit persistence coefficient, additivity across heads, and one label for both the compatibility record and the content channel; and it does not cover the normalization and MLP maps or the collapse of the predictive mixture to a point state between layers.

For fixed $q_{Q,i}$ the optimizer of Eq. (47) minimizes $\mathbb{E}_{y\sim q_{Q,i}} D_{\mathrm{KL}}\big(\beta_i \| p_i^{(\tau)}(\cdot \mid y)\big)$ with $p_i^{(\tau)}(j \mid y) \propto \pi_{ij} p_{K,j}(y)^{1/\tau}$: it is the normalized geometric mean of the pointwise source posteriors under the query law. The posterior-predictive source probability $\mathbb{E}_{y\sim q_{Q,i}} \mathbb{P}_i(J = j \mid y)$ minimizes the reverse expected divergence. It is also the unrestricted minimizer of the KL to $\mathbb{P}_i$ over joints whose y-marginal is clamped to $q_{Q,i}$, and it generally differs from the Gibbs row, as for the readout of Appendix D. Apart from a row constant, the query covariance enters D_{ij} only through the trace term of Eq. (30). That term penalizes sources whose key laws are precise along the uncertain directions of the query, and it drops out when key covariances are shared. A broad query law therefore does not flatten Gibbs routing.

Remark 9 (Noisy-query routing) *A rule under which query uncertainty does flatten routing requires a different local model, in which the projected query mean is data. Under source j, let $y \sim p_{K,j}$ and let the observed query mean be $a_i = y + \varepsilon$ with independent noise $\varepsilon \sim \mathcal{N}(0, V_{Q,i})$. Then $p(a_i \mid J = j) = \mathbb{E}_{y\sim q_{Q,i}}[p_{K,j}(y)] = \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})$, the expected likelihood kernel of the two laws (Jebara et al., 2004, Secs. 2.1 and 3.1), and Bayes' rule gives the exact posterior*

$$P(J = j \mid a_i) = \frac{\pi_{ij} \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})}{\sum_{k\in S_i} \pi_{ik} \mathcal{N}(a_i; b_k, V_{Q,i} + V_{K,k})}. \tag{51}$$

At unit temperature, Gibbs routing normalizes $\pi_{ij} \exp \mathbb{E}_{q_{Q,i}}[\log p_{K,j}]$, whereas Eq. (51) normalizes $\pi_{ij} \mathbb{E}_{q_{Q,i}}[p_{K,j}]$; in the point-query limit both give the mixture responsibility of Remark 6. With shared key covariance V_*, Eq. (51) is Eq. (33) at $\tau = 1$ with V_* replaced by $V_{Q,i} + V_*$, so the query covariance enters as a receiver-dependent whitening that tempers the row. For $r = 1$, keys $b_1 = 0$ and $b_2 = 1$ with unit variance, $a_i = 0$, and a uniform prior, the unit-temperature Gibbs weight on the first source is $(1 + e^{-1/2})^{-1} \approx 0.622$ for every query variance v, whereas Eq. (51) gives $(1 + e^{-1/(2+2v)})^{-1}$, which is 0.562 at $v = 1$ and 0.511 at $v = 10$. The weights depend on the states only through projected moments, and a common invertible map of the comparison space multiplies every term by the same Jacobian factor, so they share the invariances of Theorem 2. A temperature can be introduced through the binary record of Theorem 8 after a row shift that makes the cost $-\log \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})$ nonnegative. The rule is a different routing operator rather than a reading of Eq. (12), and the evaluated model does not implement it.

5.3 Content laws and the shared-latent alternative

For one head, $z_i^{(h)}$ is exactly the mean of a declared mixture of content laws with means $T_h \xi_j$ and weights β_{ij}. A normalized local construction makes the limited probabilistic interpretation explicit. At fixed context, let $p_{K,j}(y)$ be the comparison densities of Eq. (9),

and choose normalized content laws $\nu_j(\zeta)$ with those content means. Then

$$\mathbb{P}_i(j, y, \zeta) = \pi_{ij} p_{K,j}(y) \nu_j(\zeta), \tag{52}$$

$$\mathbb{Q}_{i,\beta}(j, y, \zeta) = \beta_j q_{Q,i}(y) \nu_j(\zeta), \tag{53}$$

$$D_{\mathrm{KL}}(\mathbb{Q}_{i,\beta} \| \mathbb{P}_i) = \sum_j \beta_j D_{\mathrm{KL}}(q_{Q,i} \| p_{K,j}) + D_{\mathrm{KL}}(\beta \| \pi_i). \tag{54}$$

The identity follows by cancellation of ν_j in the density ratio and integration of normalized factors. It recovers the uncapped routing objective at $\tau_h = 1$. Since every normalized ν_j cancels, this construction permits a chosen content bank but does not select its means, T_h, O_h, or the residual. At general τ_h, Eq. (11) is the declared entropy-regularized extension, not the KL of this same joint without additional modeling changes; the binary compatibility record of Theorem 8 realizes it exactly at every temperature. The identity determines routing within the declared joint; a predictive architecture requires additional choices.

The independent-content construction also distinguishes two measurements of a distribution from two observations of one latent draw. If $X_j \sim \mathcal{N}(\xi_j, C_j)$, $Y = S_K X_j + \varepsilon$ with independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, R)$, $R \succeq 0$, and $V = TX_j$, then Gaussian conditioning instead gives (Petersen and Pedersen, 2012)

$$\mathbb{E}[V \mid Y = y, J = j] = T\xi_j + TC_j S_K^\top (S_K C_j S_K^\top + R)^{-1}(y - S_K \xi_j). \tag{55}$$

Thus uncorrected values $T\xi_j$ do not generally equal shared-latent conditional means. Overlapping transmitting and comparison subspaces are permitted, but they do not remove this correlation term. The mixture-mean readout in Eq. (16) specifies a different statistical operation from this shared-latent conditioning law. With $R = 0$ and the query law as the recognition of Y, this model gives the same routing objective and the pair-dependent content $T\big(\xi_j + R_{K,j}(S_Q \xi_i - S_K \xi_j)\big)$, where $R_{K,j} = C_j S_K^\top (S_K C_j S_K^\top)^{-1}$ is the key-side analogue of the lift in Eq. (64). The uncorrected value is recovered for every query exactly when $TC_j S_K^\top = 0$.

The same variational reading covers the decoder. The linear decoder of Eq. (25) is exactly the unit-temperature Gibbs readout of shared-covariance Gaussian templates; Appendix D derives this identity and develops the Gaussian-bank readout with class-specific covariances.

6 Matrix Links: Rank, Composition, and Lattice Structure

Communication uses selected directions of an agent's framed state. The mathematical meaning of using a subset of a frame is that a receiving or sending map accesses a lower-dimensional observable of the state expressed in that frame. These maps also set gains and identify transmitted directions with received directions. They need not select coordinate entries, define orthogonal projections, or select a subset of group elements. The possible analogy with information coding is an interpretation to develop; no coding theorem or channel-capacity claim is assumed here.

Proposition 10 (Rank and composition of content links) *Fix a layer and write* $M_h = O_h T_h$.

1. *Each head has rank at most s_h, and the routed link of Eq.* (20) *satisfies*

$$\operatorname{rank} \mathcal{M}_{ij} \le \min\left(K, \sum_h s_h\right). \tag{56}$$

2. *For one head, intermediate frame factors cancel:*

$$L^{(h)}_{ij} L^{(h)}_{jk} - L^{(h)}_{ik} = U_i O_h (T_h O_h - I_{s_h}) T_h U_k^{-1}, \tag{57}$$

$$L^{(h)}_{i_0 i_1} \cdots L^{(h)}_{i_{a-1} i_a} = U_{i_0} M_h^a U_{i_a}^{-1}. \tag{58}$$

If $T_h O_h = I_{s_h}$, the composition defect vanishes. Conversely, if O_h has full column rank and T_h has full row rank, a vanishing defect implies $T_h O_h = I_{s_h}$.

3. *For a declared link $\mathcal{L}_{ij} : E_j \to E_i$ of any rank and $Q_i \succ 0$, the kernel*

$$k_i \mid k_j \sim \mathcal{N}(\mathcal{L}_{ij} k_j, Q_i) \tag{59}$$

is normalized, and a source $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$ has the nondegenerate receiving marginal

$$k_i \sim \mathcal{N}\left(\mathcal{L}_{ij}\mu_j, Q_i + \mathcal{L}_{ij}\Sigma_j \mathcal{L}_{ij}^\top\right). \tag{60}$$

Proof (1) Each head contributes $U_i M_h U_j^{-1}$, of rank at most s_h, and the rank of a sum is at most the sum of the ranks. (2) Substituting Eq. (17), the factors $U_j^{-1} U_j$ cancel and leave $U_i O_h T_h O_h T_h U_k^{-1} - U_i O_h T_h U_k^{-1}$; the path identity follows by induction. For the converse, remove the invertible frames and multiply the defect by a left inverse of O_h on the left and a right inverse of T_h on the right. (3) Write $k_i = \mathcal{L}_{ij} k_j + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, Q_i)$. Means and covariances give the marginal, and $Q_i \succ 0$ makes it nondegenerate. These are affine-Gaussian identities (Bishop, 2006). ■

For example, at $K = 2$, take $O_1 T_1 = \operatorname{diag}(1, 0)$ and $O_2 T_2 = \operatorname{diag}(0, 1)$. Positive weights produce the invertible reference operator $\operatorname{diag}(\beta_{ij}^{(1)}, \beta_{ij}^{(2)})$, so rank-deficient heads can sum to an invertible link. Conversely, a single head with $s_h < K$ cannot be a $\mathrm{GL}(K)$ element. The realized matrices also remain in the span of the shared $O_h T_h$ at a fixed layer; they are not unrestricted independent edge parameters.

The composition defect measures repeated receiving-to-sending processing. If $T_h O_h = I_{s_h}$, then $L^{(h)}_{ij}$ identifies $U_j \operatorname{im} O_h$ with $U_i \operatorname{im} O_h$ and composes flatly on these subspaces; its ambient self-map is a projector. At $K = s_h = 1$, $U_i = 1$, $T_h = 1$, and $O_h = 2$, the one-edge map is 2 while the two-edge map is 4. Even invertibility therefore does not make the shared processing family a flat reference transport. Paths of the same length and endpoints still agree for this unweighted, shared-head specialization. Routed mixtures can depend on the intermediate sites through their coefficients. Without the rank assumptions the converse can fail.

Under a passive frame change, $Q_i' = g_i Q_i g_i^\top$ together with the link endpoint law gives the corresponding pushforward conditional and marginal. This channel provides a probabilistic use of the sending and receiving map, but it does not force a particular predictive architecture. In particular, a residual update of the mean with covariance retained as a routing

state does not propagate the covariance in Eq. (60); Theorem 8 gives the moment-matched alternative for the full multi-head update. A population model requires a normalized directed assembly of kernels, or a separately normalized cyclic law. Multiplying reciprocal conditionals without that construction does not specify a joint evidence model.

6.1 Comparison maps and induced links

The query and key maps first define the comparison cospan

$$E_i \xrightarrow{A_{Q,i,h}} H_h^{\mathrm{cmp}} \xleftarrow{A_{K,j,h}} E_j. \tag{61}$$

The Gaussian observables of Eqs. (8) and (9) occupy this common comparison space. A cospan compares two observables; it does not by itself choose a map back from the comparison space to E_i.

The distinction can be seen directly in the mean cross term of the Gaussian KL. Suppress h, and put $P_{K,j} = (A_{K,j}\Sigma_j A_{K,j}^\top)^{-1}$. The comparison tensor is

$$B_{ij} = A_{Q,i}^\top P_{K,j} A_{K,j}, \qquad B'_{ij} = g_i^{-\top} B_{ij} g_j^{-1}. \tag{62}$$

It maps E_j to E_i^*, with invariant scalar $\mu_i^\top B_{ij}\mu_j$. A vector link instead maps E_j to E_i. A declared receiver metric can raise the first index: $G_i^{-1}B_{ij}$ then obeys the link law because $G'_i = g_i^{-\top} G_i g_i^{-1}$. The covariance Σ_i, viewed as a map $E_i^* \to E_i$, supplies another such choice; with the Fisher metric $G_i = \Sigma_i^{-1}$ the two choices coincide. Neither choice is an inverse of the rectangular query map.

A more specific lift uses the receiver's Gaussian geometry. Define

$$R_{Q,i} = C_i S_Q^\top (S_Q C_i S_Q^\top)^{-1}, \qquad S_Q R_{Q,i} = I_{r_h}, \tag{63}$$

$$L_{ij}^{\mathrm{cmp}} = U_i R_{Q,i} S_K U_j^{-1}. \tag{64}$$

For every $y \in \mathbb{R}^{r_h}$, $R_{Q,i}y$ is the unique minimizer of $\delta^\top C_i^{-1}\delta$ subject to $S_Q\delta = y$. To check this, write $\delta = R_{Q,i}y + w$ with $S_Q w = 0$. Then $(R_{Q,i}y)^\top C_i^{-1} w = y^\top (S_Q C_i S_Q^\top)^{-1} S_Q w = 0$, so the objective is its value at $R_{Q,i}y$ plus $w^\top C_i^{-1} w$. This also identifies the lift in the Gaussian conditional mean,

$$\mathbb{E}[X_i \mid S_Q X_i = y] = \xi_i + R_{Q,i}(y - S_Q\xi_i), \quad X_i \sim \mathcal{N}(\xi_i, C_i). \tag{65}$$

For the conditional identity, the Gaussian residual $X_i - \xi_i - R_{Q,i}(S_Q X_i - S_Q\xi_i)$ has zero cross covariance with $S_Q X_i$, since $C_i S_Q^\top - R_{Q,i} S_Q C_i S_Q^\top = 0$. Its joint Gaussian characteristic function therefore factors, making the residual independent of the measurement and giving Eq. (65). Thus L_{ij}^{cmp} is a precise state-dependent comparison link induced by a specified lift. Its covariance follows because $R_{Q,i}$ is unchanged by the passive site action. It transports a comparison displacement into the receiver; the complete conditional mean includes the affine centering term in Eq. (65). This construction gives S_Q and S_K a precise induced-link interpretation. It is distinct from the independently chosen content link $L_{ij}^{(h)}$ built from T_h, O_h; equality would require additional conditions on those maps.

6.2 Flat reference identification, loops, and lattice structure

The separate reference identification is

$$\Omega_{ij} = U_i U_j^{-1}, \qquad k_j \mapsto \Omega_{ij} k_j. \tag{66}$$

It is recovered from a content link when $O_h T_h = I_K$. When the frames are each represented by one exponential, it equals $e^{\phi_i} e^{-\phi_j}$; the chart restrictions of Section 2.1 remain applicable.

Remark 11 (Flat reference identification) *For any invertible frames and any finite sequence of sites, successive factors cancel, $\Omega_{ij}\Omega_{jk} = U_i U_k^{-1} = \Omega_{ik}$. The ordered product of the Ω maps therefore depends only on its endpoints, and every closed product equals I_K; in particular $\Omega_{ij}\Omega_{jk}\Omega_{ki} = I_K$. The reference identification is pure gauge: it is the coboundary of the frame field.*

The global reference space supplies this trivialization explicitly. A finite discrete base has no overlap or continuum topology obstruction here; the graph may nevertheless support independent connection data with nonidentity cycle products. The identity just proved applies to Ω, not to every matrix communication link.

Theorem 2 records the loop invariants of routed links; the following definition applies to any endpoint-covariant edge matrices. For a declared closed walk $\gamma = (i_0, \dots, i_r = i_0)$ and endpoint-covariant edge matrices $L_{ij} : E_j \to E_i$, define

$$\mathcal{H}_\gamma(L) = L_{i_0 i_1} \cdots L_{i_{r-1} i_0}, \tag{67}$$

$$\mathcal{H}_\gamma(L)' = g_{i_0} \mathcal{H}_\gamma(L) g_{i_0}^{-1}, \tag{68}$$

$$\mathcal{W}_\gamma(L) = \operatorname{Tr} \mathcal{H}_\gamma(L). \tag{69}$$

Intermediate gauge factors cancel, and cyclicity of trace proves that $\mathcal{W}_\gamma$ is invariant even for singular links. It is not a complete invariant of the loop matrix. For routed links, write $M_{ij} = \sum_h \beta_{ij}^{(h)} M_h$. Then

$$\mathcal{H}_\gamma(\mathcal{M}) = U_{i_0} \big(M_{i_0 i_1} \cdots M_{i_{r-1} i_0} \big) U_{i_0}^{-1}. \tag{70}$$

Expanding the product gives the loop-trace polynomial of Theorem 2. A nonidentity communication product can express gain, contraction, or rank loss and does not imply nonzero holonomy of Ω. Causal support has no directed cycles through distinct token positions, so on it the only closed walks are self-loops, whose traces are those of the frame-free operator M_{ii}. A loop observable therefore requires a declared extended graph and its additional links; causal routing weights alone do not provide reverse transports. Products at different network depths are compositions of different operators and are not automatically loops of one fixed connection.

A group-valued connection instead supplies invertible edge maps Θ_{ij} together with inverse reverse maps $\Theta_{ji} = \Theta_{ij}^{-1}$ (Wilson, 1974). A factorized content head decomposes exactly as

$$L_{ij}^{(h)} = F_{i,h} \Omega_{ij}, \qquad F_{i,h} = U_i O_h T_h U_i^{-1}. \tag{71}$$

Replacing Ω_{ij} by an independently specified Θ_{ij} would define $F_{i,h}\Theta_{ij}$ with the same endpoint covariance. Such a connection may have nontrivial holonomy. It is a broader construction

than the factorized link family in Eq. (17). An invertible routed link can be supplied with an inverse reverse edge, but a separately computed reverse attention operator need not be that inverse. General matrix links have a precedent in linear lattice gauge theory (Wetterich, 2014); its compact-group action and continuum conclusions are not consequences of the present algebra. Because the attention links are composites of site data rather than independent edge variables, the closer counterpart is a cellular sheaf, whose restriction maps into an edge stalk play the role of the comparison cospan of Eq. (61) (Hansen and Ghrist, 2019; Bodnar et al., 2022).

Finally, arbitrary invertible coordinate changes must be distinguished from symmetries of a fixed numerical metric or a restricted frame parameterization. Here G_i and all Gaussian tensors transform together. Holding G_i fixed restricts metric-preserving active maps to its orthogonal group, and the Euclidean quantity $\mathrm{Tr}(L_{ij}^{\top} L_{ij})$ is not invariant under general independent $\mathrm{GL}(K)$ actions. The results above establish covariance of the full-Gaussian communication construction. They do not supply an invariant lattice action or a normalized population model.

7 Language-Model Realization and Evaluation

The following experiment evaluates the causal residual architecture of Sections 2.3 and 2.4. Its saved configuration and evaluation protocol define the empirical claim. The theoretical results apply to full Gaussian states; the stored covariance representation used in this experiment is stated below.

7.1 Forward computation

The software selectors for the evaluated path are `projected_filtering` and `predictive_content`. The former names projected Gaussian routing; the latter selects direct value/output aggregation. This combination performs no iterative belief update. The route requires `gradient_mode=filtering` and inclusion of attention entropy for admission; neither setting is read by the predictive map, which performs no inner descent, so all source, score, and content dependencies remain available to outer cross-entropy differentiation.

Algorithm 1 specifies the forward and training operations. For the reported experiment, $K = 256$, $H = B = 8$, $d_b = r_h = s_h = 32$, $L = 4$, and $\tau_h = \sqrt{32}$. The implementation fixes $H = B$ and $r_h = s_h = d_b$ and computes $\tau_h = \kappa_\beta \sqrt{d_b}$ with $\kappa_\beta = 1$, so $\tau_h = \sqrt{r_h}$ holds by construction; independent head counts and ranks are not implemented. The numerical KL cost is bounded to $[0, 8K]$ before routing. The categorical optimization remains exact for this bounded cost; its identification with the unmodified Gaussian KL requires an inactive bound. Capping was checked on one 128-token window at each of 200 logged training steps and 10 evaluations, with no capped score, and on sampled windows the largest causal cost is about 300, well below $8K = 2048$. The bounded cost is nonnegative, so the evaluated routing is also the exact label posterior of the binary-record joint of Theorem 8 for that cost. The theoretical comparison in Section 4 is stated for uncapped costs.

Algorithm 1 Causal Gaussian attention and supervised predictive learning

Require: Input tokens $x_0, \dots, x_{N-1}$ and, during training, targets $y_i = x_{i+1}$ for $0 \le i < N$, vocabulary Gaussian and frame tables, layer maps $S_{Q,h}^{(\ell)}, S_{K,h}^{(\ell)}, T_h^{(\ell)}, O_h^{(\ell)}$, and decoder parameters.

Initialize $(\mu_i^{(0)}, \Sigma_i^{(0)})$ from the vocabulary prior indexed by x_i; realize the ordered token–position frame $U_i = \exp(X_{x_i}) \exp(Y_i)$.
Set $\log \pi_{ij}^{(h)} = -\alpha_h (i - j)$ up to a row constant for $j \le i$ and $\pi_{ij}^{(h)} = 0$ otherwise, with positive headwise slopes α_h and temperatures τ_h.
for $\ell = 0, \dots, L-1$ **do**
  Form $\xi_i = U_i^{-1} \mu_i^{(\ell)}$ and $C_i = U_i^{-1} \Sigma_i^{(\ell)} U_i^{-\top}$.
  for $h = 1, \dots, H$ **do**
    Form the query and key Gaussian pushforwards under $S_{Q,h}^{(\ell)}$ and $S_{K,h}^{(\ell)}$; denote their KL divergence by $D_{ij}^{(\ell,h)}$.
    Apply the numerical cost bound stated in the experimental methods to obtain $\bar{D}_{ij}^{(\ell,h)}$, then set $\beta_{ij}^{(\ell,h)} = \mathrm{softmax}_j\big(\log \pi_{ij}^{(h)} - \bar{D}_{ij}^{(\ell,h)} / \tau_h\big)$.
    Compute $z_i^{(\ell,h)} = \sum_j \beta_{ij}^{(\ell,h)} T_h^{(\ell)} \xi_j$.
  end for
  Compute $m_i^{(\ell)} = U_i \sum_h O_h^{(\ell)} z_i^{(\ell,h)}$ and $\mu_i^{\mathrm{att},(\ell)} = \mu_i^{(\ell)} + m_i^{(\ell)}$.
  Set $u_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}(\mu_i^{\mathrm{att},(\ell)})$.
  Set $\mu_i^{(\ell+1)} = u_i^{(\ell)} + W_2^{(\ell)} \,\mathrm{GELU}(W_1^{(\ell)} u_i^{(\ell)} + b_1^{(\ell)}) + b_2^{(\ell)}$ and $\Sigma_i^{(\ell+1)} = \Sigma_i^{(\ell)}$.
end for
Compute $p(y_i \mid x_{\le i}) = \mathrm{softmax}\big(W_{\mathrm{out}} \mu_i^{(L)} + b_{\mathrm{out}} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}}\big)_{y_i}$.
if training **then**
  Backpropagate the mean next-token cross-entropy through all layers, routing weights, content maps, frames, and initial vocabulary parameters; update trainable parameters with AdamW (Loshchilov and Hutter, 2019).
end if

7.2 Architecture and training

The WikiText-103 (Merity et al., 2017) experiment instantiates Section 2.3 with $K = 256$, eight 32-dimensional gauge blocks, four layers, and seed 6. Each layer has independent comparison and content maps. The configuration uses `projected_filtering` with full input scope, `predictive_content` readout, full output scope, and the additive attention residual. The group parameterization remains block-diagonal and vertex-factorized, with learned token and positional frames. Each block also contains affine LayerNorm and a coordinate GELU MLP with hidden width 2048, expansion factor eight, zero dropout, and covariance passthrough. These are the combined methods used in the recorded run; the saved results do not isolate the contribution of each choice.

The stored beliefs are diagonal Gaussians, but each role projection retains its full 32×32 reduced covariance. Diagonal storage is a partial gauge fixing: within each frame block every positive-definite covariance is congruent to a diagonal one and the frame absorbs the congruence, so with block-diagonal frames the realized reference covariances C_i are exactly

the block-diagonal positive-definite matrices, and cross-block reference covariance is not representable. The operative router is Gaussian KL. Although the launch configuration contains `attention_score=squared_distance`, `mean_only=False` selects the Gaussian-KL path, as the saved resolved-operator metadata confirms. The causal ALiBi prior includes the current position and adds the headwise logit bias $-2^{-h}(i-j)$ for $j \le i$, $h = 1, \dots, 8$. The routing temperature is $\sqrt{32}$. Predictive content uses one readout per layer; the configured inference learning rates and belief trust bounds do not scale this update. The covariance and frame are held fixed through the stack, while outer cross-entropy training updates the learned tables, frames, role maps, MLPs, normalization, and decoder parameters through the forward graph. The decoder term $\kappa_{\text{uni}} \log p_{\text{uni}}$ uses the fixed coefficient $\kappa_{\text{uni}} = 1$ and the add-one-smoothed unigram frequency of the training stream; because b_{out} is learned without decay, this term acts as an initialization offset for b_{out}.

The stored covariance table holds log-variances initialized at $\log 15$, so the initial variance is 15. Decoupled weight decay 0.04 on this table carried the variances to 1.045 ± 0.054 per coordinate at the final step, with 84.6% of entries within 10% of unit variance; the run's covariance-diversity check recorded a coefficient of variation of $\operatorname{Tr}\Sigma$ of 0.047 against its threshold of 0.10. Because covariances and frames are fixed through the stack, $C_i = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}$ depends only on token type and position and is close to $1.04(U_i^\top U_i)^{-1}$, with a median relative Frobenius gap of 7.6% on sampled token-position pairs: the covariance channel of the router is carried by the frames. On sampled windows of 128 tokens, replacing every routing covariance C_i by the identity raises cross-entropy by 0.325 nats on average over 8 windows (range 0.11 to 0.57), whereas replacing only the stored variances by the identity raises it by 0.024 ± 0.024 nats over 24 windows. The decay also acted as a hidden temperature for the Mahalanobis term, which a common rescaling of all covariances by c rescales by $1/c$; over training it sharpened that term about 14-fold. The experiment therefore does not test learned per-token uncertainty.

The run records 456,654,162 parameters, including 1,048,576 relational-map parameters and 4,203,520 MLP parameters. The token-frame table has $50{,}257 \times 8{,}192 = 411{,}705{,}344$ parameters, approximately 90.16% of the total. Per-type tables (token frames, prior means and log-variances, and decoder weights and biases) hold 98.6% of the parameters, the positional frames hold 1,048,576, and the context-processing parameters (interface maps, MLPs, and the shared normalization) number 5,252,608. This is capacity accounting; sparse table access does not make the count equivalent to dense-transformer FLOPs. The run completed 240,000 accepted updates and 491,520,000 training-token presentations over about 116.8 million unique training tokens, or about 4.21 passes. The saved environment identifies an NVIDIA GeForce RTX 5090, PyTorch `2.10.0.dev20251210+cu128`, and bfloat16 automatic mixed precision, which applies to the MLP matrix products; frame exponentials, Gaussian scores, content aggregation, layer normalization ($\epsilon = 10^{-6}$), decoding, and cross-entropy run in float32, and GELU is the exact error-function form.

Evaluation uses context length 128 and stride 32, excluding overlapping positions from repeated scoring. The test result covers 280,541 next-token targets from 280,542 tokens; validation covers 245,020 targets. The cache filename identifies tiktoken/GPT-2 tokenization with vocabulary size 50,257, but no verified tokenizer manifest is available. Perplexities refer to this saved protocol and are not directly comparable to conventional word-level WikiText-103 results. Table 3 lists the recorded perplexities.

Evaluation	Step	Perplexity
Best recorded validation	216,000	29.6088
Final validation	240,000	30.0100
Test, final weights	240,000	31.0389
Test, best-validation weights	216,000	pending
Training-split sample, evaluation protocol	240,000	pending

Table 3: Single-seed Gaussian attention language-model experiment. Test evaluation used the final weights, not the checkpoint with the lowest recorded validation perplexity. No variability estimate or causal ablation is supplied by this run.

The test cross-entropy is 3.43524149 nats per scored token. Validation cross-entropy reached its minimum at step 216,000 and rose by 0.0135 nats by the final step; the test split has not yet been evaluated at the best-validation checkpoint. The result is stored in the run directory `31.04_wikitext-103_K256_4-layer_noHM`, with configuration and resolved methods in `config.json`, evaluation in `test_results.json`, and source/data identity in `provenance.json`. The provenance records a dirty source tree and a change in source identity during the run. The starting Git revision is `4e137ec88f45` and the final recorded revision is `965718d88cf7`; these revisions alone cannot identify the executed dirty-source bytes. The result therefore has incomplete source reproducibility.

A saved intervention probe on the first validation batch, 16 overlapping windows that score 608 consecutive targets, reports cross-entropy 3.0431 for the trained route (against 3.4015 on the full validation split), 4.1923 when the routing energy is removed, and 5.1182 when the positional prior is flattened on causal support. These frozen-model batch interventions support reliance on both content-dependent routing and the positional prior in that batch. They are not corpus-wide test results, retraining ablations, or evidence that any individual projection, the residual, or the MLP is necessary. These interventions measure prediction error, not variational descent.

In the trained model the explicit attention residual is numerically inert. On 24 sampled windows of 128 tokens, the per-token ratio $\|m_i\|/\|\mu_i\|$ has median 2.1×10^4, 5.9×10^3, 5.3×10^3, and 1.8×10^3 at layers 0 to 3, with minimum 693. The residual changes the normalized state by at most 1.8×10^{-3} in relative terms, and deleting it at every layer changes cross-entropy by less than 4×10^{-4} nats on every window. The relational maps train without weight decay and the layer normalization is invariant to the scale of its input, so the largest singular values of $O_h T_h$ grew from 1 at initialization to between 2.6×10^3 and 2.6×10^4 across heads and layers. At the final weights each layer therefore acts as $u_i = \mathrm{LN}(m_i)$ up to a relative 2×10^{-3}. The receiving token reaches the next layer through its frame and its routing self-weight, whose mean lies between 0.07 and 0.15 across layers and windows, rather than through the additive state. The role of the residual earlier in training was not measured.

Saved per-layer probes from one 128-token test window report minimum raw comparison-map singular values from 6.64 to 12.32, maximum projected-covariance condition numbers from approximately 88.7 to 550.5, all attained by query covariances,

Frozen intervention on every validation target	CE	ΔCE	95% interval
None (final weights)	pending	0	pending
Routing costs removed; positional prior retained	pending	pending	pending
Positional prior flattened on the causal support	pending	pending	pending
Residual term removed at every layer, $\mu_i + m_i \mapsto m_i$	pending	pending	pending
Stored covariances $\Sigma_{p,v}$ replaced by the identity	pending	pending	pending
All routing covariances C_i replaced by the identity	pending	pending	pending
Token frames replaced by the identity in every role	pending	pending	pending
Positional frames replaced by the identity in every role	pending	pending	pending
Frames removed from the input moments only	pending	pending	pending
Frames removed from the routing covariance only	pending	pending	pending
Frames removed from the normalization and MLP conjugation only	pending	pending	pending
Frames removed from the decoder only	pending	pending	pending

Table 4: Reserved frozen interventions on the complete validation split. Each row modifies one input of the trained predictor and records the change in cross-entropy, with intervals from an article-level bootstrap. The role-wise frame rows follow Figure 2. These are properties of one trained model; they are not retraining ablations and do not establish that any component is necessary.

and zero recorded covariance repairs or capped scores. Both describe the numerical representative reached by the optimizer rather than the learned routing. The redundancy $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ of Theorem 2 changes them while leaving every cost unchanged, and the raw maps, which carry no weight decay, grew by factors of about 40 to 54 in norm along this direction during training. The basis-independent quantities are the generalized eigenvalues of the pairs $(V_{Q,i}, V_{K,j})$ and the relative gains, the square roots of the eigenvalues of $(S_{K,h}S_{K,h}^\top)^{-1}S_{Q,h}S_{Q,h}^\top$, which lie between 0.11 and 5.35. The mathematical full-row-rank condition does not itself guarantee conditioning throughout learned optimization.

The result supports feasibility of the combined architecture under the stated protocol.

8 Discussion

8.1 Contribution and limits

The mathematical contribution is a specified full-Gaussian routing and predictive-content construction, together with its factorization into covariant matrix links. Reference coordinates permit full-input role measurements, exact reduced Gaussian KL comparisons, and a content channel with an explicit passive covariance law. Separate comparison and content maps are familiar from standard attention; their roles here are defined inside the framed Gaussian model. The independent-content joint establishes a consistent interpretation while leaving the content maps underdetermined. It therefore cannot, by itself, explain their empirical success or claim priority for Gaussian-KL or gauge-equivariant attention.

PLACEHOLDER M4

Panel A: for each layer and head, the variance of each routing-logit component and its covariance with the total logit: whitened cross term $\mathsf{q}_i^\top \mathsf{k}_j/\tau_h$, key-norm term, source-dependent trace and log-determinant terms of Eq. (30), and positional prior.

Panel B: after whitening by the pooled projected key covariance, the distribution of ε_i from Eq. (37) and of the rowwise ℓ_1 distance between Gaussian and dot-product routing, with the bound of Eq. (38).

Panel C: dispersion of the projected key covariances $V_{K,j}$ across sources, and the share of that dispersion induced by the frames rather than by the stored covariances.

TO COLLECT: validation-set score components or projected moments for all layers and heads, with sampling coverage.

Figure 3: Reserved diagnostics for the conditions of Proposition 5 in the trained model. Panel A attributes the routing logits to their Gaussian components; Panels B and C measure the distance of the trained model from the constant-norm and shared-covariance hypotheses. The figure characterizes the realized operator. Whether covariance-aware routing improves prediction requires the retraining comparisons of Table 5.

PLACEHOLDER M3

Panel A: held-out cross-entropy versus training-token presentations for every arm of Table 5, one curve per seed.

Panel B: the same curves against measured device-hours.

Panel C: held-out minus training cross-entropy, with the training term evaluated on a fixed training-split sample under the evaluation protocol.

TO COLLECT: per-seed validation curves at a common cadence, token counts, passes over the unique training tokens, synchronized wall-clock time, and the checkpoint-selection rule.

Figure 4: Reserved learning curves for the arms of Table 5. Panels A and B separate data efficiency from compute efficiency. Panel C records the generalization gap on one protocol, which bears on interpretation because per-type tables hold most of the parameters.

Model arm	Seeds	Parameters	Test CE	Test PPL	GPU-hours
Retraining arms at the evaluated configuration and token budget					
Evaluated: learned frames, decayed stored covariance	pending	pending	pending	pending	pending
Learned frames, stored covariance without decay	pending	pending	pending	pending	pending
Identity frames, decayed stored covariance	pending	pending	pending	pending	pending
Identity frames, stored covariance without decay	pending	pending	pending	pending	pending
Identity frames, per-type additive table of equal size	pending	pending	pending	pending	pending
Attention residual removed	pending	pending	pending	pending	pending
LayerNorm and MLP in reference coordinates	pending	pending	pending	pending	pending
Intertwiner-constrained nonlinear block	pending	pending	pending	pending	pending
MLP removed	pending	pending	pending	pending	pending
MLP replaced by Gaussian template routing	pending	pending	pending	pending	pending
Conventional transformers: same tokenizer, context, stride, and budget					
Matched context-processing parameters	pending	pending	pending	pending	pending
Matched measured compute	pending	pending	pending	pending	pending
Per-type table of the frame table's size	pending	pending	pending	pending	pending
Word-level protocol, every target scored					
Gaussian attention	pending	pending	pending	pending	pending
Conventional transformer, matched	pending	pending	pending	pending	pending

Table 5: Reserved main results under one declared protocol. Each retraining arm changes one component of the evaluated realization; the two reference-coordinate and intertwiner arms make the complete stack frame-covariant. Report the mean and standard deviation over at least three replicates that vary both the model seed and the data-order seed, paired across arms, with per-seed values in Table 8. Give parameters as context-processing (non-embedding) and total counts, since per-type tables hold most of the storage. Word-level rows permit comparison with published WikiText-103 results and are not compared with the BPE rows. Cells remain pending until each evaluation is complete.

The covariance of Theorem 2 holds for any operator that reads the states through reference moments, so it does not by itself constrain the learned maps; the coordinate normalization, MLP, and decoder are where the learned frames act (Section 3). Covariance passthrough is not propagation of predictive uncertainty; Eq. (50) gives the moment-matched alternative. The source-selection objective is exact for its declared costs, and routing and content have exact probabilistic readings, but no common-free-energy descent statement is established for layer composition.

The single recorded run supplies no matched baseline or variability estimate. It cannot establish parameter efficiency, sample efficiency, benefit from learned frames, or causal necessity of covariance, the residual, or the MLP. The frame table dominates parameter count, and the short-context, tokenizer-specific evaluation must be kept explicit in comparisons. Dirty and changing source provenance limits reconstruction of the executed model; a Git SHA alone does not recover its bytes.

8.2 Relation to a complete variational model

The general gauge-theoretic variational free energy (VFE) construction supplies a broader probabilistic setting in which normalized agent and observation kernels define one population joint and one exact evidence bound. Its latent-source construction yields categorical Gibbs routing under explicit likelihood and recognition restrictions; Appendix E states this connection. The predictor already has a normalized autoregressive likelihood. For the attention sublayer, a normalized latent model exists in which routing is the exact label posterior and the content-plus-residual update is the exact predictive mean (Theorem 8), and the linear decoder is a Gibbs readout (Section 5.3). Unresolved are the normalization and MLP maps, the collapse of the predictive mixture to a point state between layers, and the relation of cross-entropy training to that model's evidence. This distinction preserves the exact attention result without attributing the whole architecture to the Free Energy Principle.

8.3 Selective communication and information coding

The sending and receiving interfaces describe selected use of a framed state. They impose no claim that a learned direction is a measured cognitive faculty. A connection with Shannon-style coding remains an open question. The Gaussian channel of Proposition 10 provides one possible channel law; a coding formulation additionally requires a message model, an input or resource constraint, and a specified information measure. Its relation to the predictive mean operator remains to be established. Subspace dimension and low matrix rank alone do not supply a channel capacity.

8.4 Experiments needed to separate the contributions

The next comparison should freeze source bytes, tokenizer and data manifests, checkpoint selection, and evaluation masks. A conventional projected-attention baseline should share tokenizer, context, depth, MLP, training tokens, and a stated parameter and compute budget. Paired retraining comparisons can then isolate covariance-aware routing from a matched mean-only or dot-product route, and learned frames from an identity-frame model

with its parameterization change reported. The mean-only and dot-product route used for that control was removed from the reference implementation after the reported run and must be restored in a tagged release; a capacity control with an additive per-type table of equal size is implemented. Residual and MLP necessity require their own controlled comparisons. Multiple paired seeds are needed to estimate variability; the existing frozen-batch probes do not replace retraining experiments.

Block-diagonal frames and cross-block communication are distinct design choices. Full-input S_Q, S_K, T and full-output O already communicate across frame blocks when the optional cross-block head-mixing layer is disabled, as here. A future transport-group comparison should hold these communication maps fixed. Likewise, projected-key norm measurements can test the conditional dot-product reduction without assuming that LayerNorm enforces it. These are bounded tests of the attention mechanism.

9 Conclusion

Gaussian-KL routing between framed Gaussian states is the exact minimizer of an entropy-regularized source-selection objective and, at every positive temperature, the exact posterior of a source label under a normalized binary compatibility record; at unit temperature it is also the coordinate minimizer of a local Gaussian mixture KL. In dual coordinates it is a dot product between query expectation parameters and key natural parameters. With shared projected key covariance it equals whitened dot-product routing with a key-norm penalty that neither LayerNorm nor high-dimensional concentration removes; with source-dependent covariance it adds a per-source metric and pair biases. Content aggregation with its residual is the exact predictive mean of a normalized mixture-of-channels kernel. The attention operator factors through frame-invariant reference moments, so its covariance constrains no learned map, and the frames affect computation only through the state parametrization and the frame-dependent normalization, MLP, and decoder.

Sending and receiving maps factor matrix-valued links on the token graph. These links are covariant composites of site data, of the bilinear kind distinguished by Wetterich (2014, Sec. I): in reference coordinates each routed link is $M_{ij} = \sum_h \beta_{ij}^{(h)} O_h T_h$, and each loop trace is a polynomial in the routing weights. The reference transport is pure gauge, and the causal support contains no closed walk through distinct positions. An independent edge connection with nontrivial holonomy, together with an action for it, is required before the construction carries lattice-gauge degrees of freedom.

The four-layer model records final-weight test perplexity 31.04 under its saved protocol. This is a feasibility result with one seed and incomplete source reproducibility, in which the attention residual is numerically inert and the stored variances collapsed to near unity. A controlled baseline and component comparisons are still needed to establish an empirical advantage. A complete generative interpretation would require an explicit joint law, observation model, and variational update that recover the whole predictor. **[Data pending:** replace with the final matched comparison of Table 5 and the frozen interventions of Table 4.]

Code availability

The reference implementation is available at `https://github.com/cdenn016/V3_Transformer`. The mean-only and dot-product control route proposed in Section 8.4 was removed from that repository after the reported run, and the current configuration loader rejects the archived configuration; a tagged release containing the evaluated route, the control route, and a loader for the archived configuration is required. **[Data pending:** tag, commit, and archive identifier of that release.] Section 7.2 identifies the saved configuration, evaluation record, and source-provenance limits of the reported experiment. A public repository does not by itself recover the modified source bytes used in that run.

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Appendix A. The Smooth-Bundle Construction

This appendix gives the construction behind the frames of Section 2.1. Figure 5 gives the geometric picture.

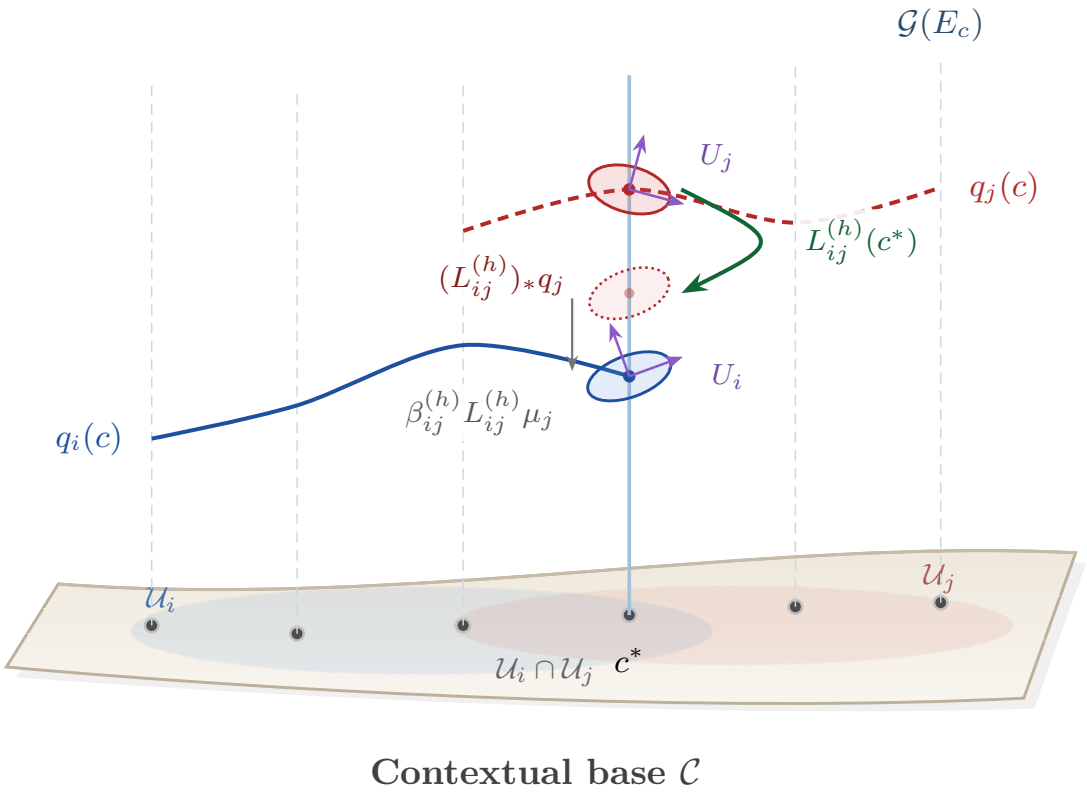


Figure 5: **Local Gaussian sections and directed communication links.** The curved surface represents the contextual base of the principal bundle; the curves are sections of its associated Gaussian-state bundle over overlapping neighborhoods. At a shared context c^*, the local frames accompany the state glyphs. For one communication head indexed by h, the green arrow applies $L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}$ to the sender. The dotted contour depicts its Gaussian pushforward; the gray arrow identifies the weighted mean contribution to the receiver. A full-rank link is illustrated; a lower-rank link may produce a singular pushforward. The receiver's state need not equal that pushforward. Routing uses the separate comparison channel in Figure 1. The finite token construction requires no smooth contextual base.

The geometric construction begins with a principal bundle of reference frames over a space of contexts. Let $\mathcal{C}$ be a smooth contextual base and take $G = \mathrm{GL}(K, \mathbb{R})$. A principal bundle

$$\pi : \mathcal{P} \longrightarrow \mathcal{C} \tag{72}$$

carries a free right action $n \mapsto n \cdot g$ that preserves π and is transitive on each fiber. Locally, $\mathcal{P}|_{\mathcal{U}} \simeq \mathcal{U} \times G$, with the group acting on the second factor. Thus a principal fiber consists of frame choices; choosing one frame identifies the fiber with G. The probabilistic states are attached to this principal structure through associated bundles (Nakahara, 2003; Frankel, 2011).

Let $\mathsf{V} = \mathbb{R}^K$ carry the defining representation of G, and let

$$\mathcal{B} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathsf{V},\ \Sigma \succ 0\}, \qquad \rho(g)\mathcal{N}(\mu, \Sigma) = \mathcal{N}(g\mu, g\Sigma g^\top). \tag{73}$$

The vector and Gaussian statistical bundles are

$$E = \mathcal{P} \times_G \mathsf{V}, \qquad \mathcal{G}(E) = \mathcal{P} \times_\rho \mathcal{B}. \tag{74}$$

For the statistical bundle, the quotient convention is

$$(n \cdot g, b) \sim (n, \rho(g)b). \tag{75}$$

Equivalently, (n, b) and $(n \cdot g, \rho(g^{-1})b)$ represent the same associated point. The analogous relation uses $v \mapsto gv$ for the vector bundle. The principal bundle carries gauge freedom, the vector bundle carries latent values, and the associated statistical bundle carries distributions over those values.

An agent over a contextual region $\mathcal{U}_i$ selects a local principal section u_i and maintains belief and prior sections:

$$\begin{aligned} u_i : \mathcal{U}_i \longrightarrow \mathcal{P}, \qquad & \pi \circ u_i = \mathrm{id}_{\mathcal{U}_i}, \\ q_i, p_i \in \Gamma\big(\mathcal{G}(E)|_{\mathcal{U}_i}\big). & \end{aligned} \tag{76}$$

Here Γ denotes smooth sections, and the belief and prior have coordinates (μ_i, Σ_i) and $(\mu_{p,i}, \Sigma_{p,i})$ in the section u_i. Communication additionally uses one reference trivialization σ_0 of $\mathcal{P}$ on its working domain. The frame field is the relative element U_i defined by $u_i = \sigma_0 \cdot U_i^{-1}$, so that $\xi_i = U_i^{-1}\mu_i$ are the σ_0-coordinates of the same vector. Identifying the fiber with its coordinates in the section u_i, written E_c, the frame is the linear isomorphism $U_i(c) : \mathsf{V} \to E_c$ that converts σ_0-coordinates into u_i-coordinates. On overlaps, agents can maintain different Gaussian sections at the same context. They are not required to be local representatives of one common belief. Figure 5 illustrates these associated statistical sections and their local frames. The attention construction uses this belief/prior sector; it does not require an additional bundle of generative-model parameters.

For token attention, the contextual dependence is evaluated at a fixed context c_0, while $i \in \mathcal{I} = \{0, \ldots, N-1\}$ labels the token agents. Keeping a separately labeled copy of the fiber for each token is the pullback along the constant map $\iota : \mathcal{I} \to \mathcal{C}$, $\iota(i) = c_0$:

$$\mathcal{P}_\mathcal{I} = \iota^*\mathcal{P}, \qquad E_\mathcal{I} = \iota^* E. \tag{77}$$

Write E_i for the vector fiber at token i. This notation distinguishes token incidence from the contextual base of the general construction. Because ι is constant, $\iota^*\mathcal{P} = \mathcal{I} \times \mathcal{P}_{c_0}$ canonically. The token construction therefore sees no connection, curvature, or topology of $\mathcal{P}$, and its gauge group is G^N, on which the contextual gauge group acts only through the diagonal. Placing tokens at distinct contexts $\iota(i) = c_i$ and realizing links by parallel transport along declared base curves would make the contextual connection enter the attention operator. The section of each token agent and the reference trivialization give $U_i : \mathsf{V} \to E_i$, and its statistical fiber is

$$\mathcal{G}(E_i) = \{\mathcal{N}(\mu, \Sigma) : \mu \in E_i,\ \Sigma \succ 0\}. \tag{78}$$

A general contextual principal bundle need not admit a global section. The factorized attention formulas use the reference trivialization σ_0 on their working domain; on the finite token pullback this trivialization is canonical. The communication graph is additional data; the matrix links on it are determined by the frames, the shared interfaces, and the routing weights. A link $L_{ij} : E_j \to E_i$ can be singular, so it is not in general a principal-bundle transition function or a group-valued parallel transport. Sections 2 and 6 construct these links from comparison and content interfaces while retaining the underlying principal-bundle structure.

Appendix B. Irreducible Frame Blocks and Communication Heads

The representation carried by a state and the organization of its communication heads are separate structures. To make their relationship precise, fix a decomposition $\mathsf{V} = \bigoplus_{b=1}^{B} \mathsf{V}_b$, with $\mathsf{V}_b \simeq \mathbb{R}^{d_b}$ and $K = \sum_b d_b$, and restrict the frame group to

$$G = \prod_{b=1}^{B} \mathrm{GL}(d_b, \mathbb{R}), \qquad \rho(g_1, \dots, g_B) = \bigoplus_{b=1}^{B} g_b, \qquad U_i = \bigoplus_{b=1}^{B} U_i^{(b)}. \tag{79}$$

Here B counts independently transforming frame blocks, while H counts communication heads. Each V_b is an irreducible real representation of G: a nonzero invariant subspace contains the orbit of any one of its nonzero vectors, and the bth general linear factor maps that vector to every other nonzero vector in V_b. Thus no proper nonzero invariant subspace exists. These representations remain inequivalent when their dimensions coincide, because the product factors act independently. Repeating the same group element on several blocks would instead give equivalent copies of one representation. Equation (79) is an explicit specialization of the full $\mathrm{GL}(K)$ action, not a theorem that an arbitrary attention architecture has this decomposition.

Let $\mathrm{pr}_b : \mathsf{V} \to \mathsf{V}_b$ and $\iota_b : \mathsf{V}_b \to \mathsf{V}$ be the coordinate projection and inclusion, and put $E_i^{(b)} = U_i \mathsf{V}_b$. A head h is local to a block $b = b(h)$ when its interface maps factor as

$$S_{Q,h} = \overline{S}_{Q,h} \,\mathrm{pr}_b, \quad S_{K,h} = \overline{S}_{K,h} \,\mathrm{pr}_b, \quad T_h = \overline{T}_h \,\mathrm{pr}_b, \quad O_h = \iota_b \overline{O}_h. \tag{80}$$

The barred maps act on the selected block; the comparison maps retain the full-row-rank requirement. The resulting $L_{ij}^{(h)} = U_i \iota_b \overline{O}_h \overline{T}_h \,\mathrm{pr}_b \, U_j^{-1}$ communicates only between the corresponding site subspaces $E_j^{(b)}$ and $E_i^{(b)}$. Taking $H = B$ and a one-to-one assignment gives a precise sense in which heads can be organized by irreducible blocks. The irreducible object is the vector carrier $E_i^{(b)}$, however, not the Gaussian density or the entire attention operation. The common reference channels remain fixed under the passive site action of Eq. (4). A proper selected row space inside the carrier is not invariant under the full defining $\mathrm{GL}(d_b)$ action. In the full-input/output construction, S_Q, S_K, T, O need not satisfy Eq. (80); a head can access several blocks. Its endpoint covariance follows from the transforming frames, without requiring the fixed reference maps to be group intertwiners.

Independent Gaussian blocks require an additional statistical assumption. If every local covariance is block diagonal, $\Sigma_i = \bigoplus_b \Sigma_i^{(b)}$, then the full Gaussian laws factor and block-

diagonal reference transport gives

$$D_{\mathrm{KL}}\big(q_i\|(\Omega_{ij})_*q_j\big)=\sum_{b=1}^{B}D_{\mathrm{KL}}\big(q_i^{(b)}\|(\Omega_{ij}^{(b)})_*q_j^{(b)}\big),\quad \Omega_{ij}^{(b)}=U_i^{(b)}(U_j^{(b)})^{-1}. \tag{81}$$

Here $q_i^{(b)}$ is the block marginal. The identity follows by writing the logarithm of the product-density ratio as a sum and integrating each normalized factor (van Erven and Harremøes, 2014, Thm. 28). Block-diagonal frames alone do not imply it. For example, $q=\mathcal{N}(0,\left(\begin{smallmatrix}1 & c\\ c & 1\end{smallmatrix}\right))$, $p=\mathcal{N}(0,I_2)$, and $0<|c|<1$ have identical one-dimensional marginals but $D_{\mathrm{KL}}(q\|p)=-\frac{1}{2}\log(1-c^2)>0$. Furthermore, separate routing distributions for the blocks are a modeling choice: softmax of the summed KL chooses one shared source and does not equal independent headwise source selection.

The frame generators have a related but different decomposition. At a block, $\mathfrak{gl}(d_b)=\mathfrak{sl}(d_b)\oplus\mathbb{R}I_{d_b}$ separates traceless generators from the central dilation generator; it is not a decomposition of the state space into two irreducible heads. At the ambient level,

$$\mathfrak{gl}(K)=\left(\bigoplus_b\mathfrak{gl}(d_b)\right)\oplus_{\mathrm{vec}}\mathfrak{m},\qquad \mathfrak{m}=\bigoplus_{b\neq c}\mathrm{Hom}(\mathsf{V}_c,\mathsf{V}_b),\qquad \dim\mathfrak{m}=K^2-\sum_b d_b^2. \tag{82}$$

The subscript emphasizes a vector-space splitting. The off-block complement is preserved by block-diagonal conjugation but is generally not a Lie subalgebra: for two scalar blocks, $[E_{12},E_{21}]=E_{11}-E_{22}$ is block diagonal. Restoring these generators enlarges the frame group; it does not add the only possible cross-block communication, since full-width interface maps already provide such communication.

Representation-based attention has established precedents. The SE(3)-Transformer uses irreducible rotation types with invariant attention and equivariant values; its multi-head construction partitions multiplicity channels while retaining several representation types in each head (Fuchs et al., 2020, Secs. 2.3 and 3.2, App. D.3). Gauge Equivariant Transformer combines local-frame-equivariant multi-head attention with regular fields and their irreducible decomposition (He et al., 2021, Secs. 4.1–4.2). The subject here is the particular combination of full Gaussian comparison laws, separately specified sending/receiving interfaces, and the matrix links $U_iO_hT_hU_j^{-1}$. The frame blocks specify the state representation, while the comparison and content interfaces specify how that state is used for communication.

Appendix C. Positional and Masking Priors

With a nonuniform prior, Eq. (47) adds $\log\pi_{ij}$ to the logits with unit coefficient while the cost is divided by τ, so temperature never rescales a positional preference. Masks and positional biases therefore enter as priors over sources. This identification, and the entropy-regularized or transport form of the softmax that underlies it, is established in the attention literature (Niculae and Blondel, 2017; Litman, 2025; Bianchessi et al., 2026; Litman and Guo, 2026). On the causal support $S_i=\{0,\dots,i\}$, the uniform prior $\pi_{ij}=\mathbf{1}[j\le i]/(i+1)$ gives the support and row normalization of GPT-family attention (Radford et al., 2019). A window of radius w, $\pi_{ij}\propto\mathbf{1}[i-w\le j\le i]$, gives the sliding-window support of Longformer

and Mistral (Beltagy et al., 2020; Jiang et al., 2023). The prior $\pi_{ij} \propto \exp[-\alpha_h(i-j)]$ gives the linear biases of attention with linear biases (ALiBi) (Press et al., 2022), and $\log \pi_{ij} = f_h(i-j) + \text{const}$ gives learned relative biases, parameterized by sinusoids in the global positional bias of Transformer-XL and shared within logarithmic offset buckets in T5 (Dai et al., 2019; Raffel et al., 2020). Press et al. (2022) state the geometric slope schedule $\alpha_h = 2^{-8h/H}$; their reference implementation uses it when H is a power of two and otherwise appends alternate slopes of the next power-of-two schedule. The construction fixes only the form in which a prior enters. It does not select a prior, except that the autoregressive factorization requires $\pi_{ik} = 0$ for $k > i$ at every layer, because each state depends on its own token.

Appendix D. An Optional Prior-Bank Vocabulary Readout

Let q_i be the Gaussian state after the chosen forward computation and p_v a normalized class template on the same latent space, with finite $D_{\mathrm{KL}}(q_i \| p_v)$. Declaring the templates in reference coordinates and comparing them with $\widetilde{q}_i = (U_i^{-1})_* q_i$ makes the readout invariant under the passive action; comparing local coordinates directly defines a different, non-covariant readout. No convergence of q_i is assumed. Specify a categorical readout $\hat{p}_i$ by minimizing

$$\mathcal{J}_{\mathrm{decode}}(\hat{p}_i) = \sum_v \hat{p}_i(v) D_{\mathrm{KL}}(q_i \| p_v) + \tau_{\mathrm{dec}} D_{\mathrm{KL}}(\hat{p}_i \| \omega), \qquad \tau_{\mathrm{dec}} > 0, \tag{83}$$

where ω is a class prior. The unique optimum on its support is

$$\hat{p}_i(v) \propto \omega_v \exp\left[-D_{\mathrm{KL}}(q_i \| p_v)/\tau_{\mathrm{dec}}\right]. \tag{84}$$

This is a specified Gibbs readout. At unit temperature it is also the categorical mean-field coordinate update for a fixed q_i under the normalized mixture $p(v, x) = \omega_v p_v(x)$, as in the general expected-log update of Blei et al. (2017, §2.4, Eqs. 17–18). It is not generically the posterior-predictive average of a classifier.

The score identity is exact:

$$\mathbb{E}_{q_i}[\log p_v(X)] = -D_{\mathrm{KL}}(q_i \| p_v) - H(q_i). \tag{85}$$

The entropy is independent of v and cancels in the categorical softmax. For Gaussian $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and $p_v = \mathcal{N}(\mu_{p,v}, \Sigma_{p,v})$, the expectation is

$$-\frac{1}{2}\left[\mathrm{Tr}(\Sigma_{p,v}^{-1}\Sigma_i) + (\mu_i - \mu_{p,v})^\top \Sigma_{p,v}^{-1}(\mu_i - \mu_{p,v}) + \log\det \Sigma_{p,v} + K\log(2\pi)\right]. \tag{86}$$

These are Gaussian cross-entropy identities (Bishop, 2006).

For the normalized point classifier

$$p(v \mid x) = \frac{\omega_v p_v(x)}{\sum_w \omega_w p_w(x)}, \tag{87}$$

posterior prediction instead averages probabilities, $\bar{p}_i(v) = \mathbb{E}_{q_i}[p(v \mid X)]$. This generally differs from normalizing $\exp(\mathbb{E}_{q_i} \log p(v \mid X))$. For example, with equal class priors, $q_i =$

$\mathcal{N}(0,4)$ and templates $p_0 = \mathcal{N}(0,1)$, $p_1 = \mathcal{N}(2,1)$, the unit-temperature Gibbs readout is $\hat{p}_i(1) = 1/(1+e^2) \approx 0.11920$. The predictive average is

$$\bar{p}_i(1) = \mathbb{E}_{X\sim\mathcal{N}(0,4)}\left[\frac{1}{1+\exp(2-2X)}\right] \approx 0.32383. \tag{88}$$

The expected-log construction therefore does not replace the predictive integral. Both readouts are variational: among joints whose latent marginal is fixed to q_i, the unrestricted minimizer of the KL to $\omega_v p_v(x)$ has class marginal $\bar{p}_i$, and the minimizer within the product family is the Gibbs readout. The same distinction separates the Gibbs routing row from the posterior-predictive source probability of Section 5.2.

Uniform ω gives logits $-D_{\mathrm{KL}}(q_i\|p_v)/\tau_{\mathrm{dec}}$. Projecting a full-covariance belief before decoding makes this identity exact for the projected distribution only. Choosing a temperature or learning an additional scalar is a modeling choice. Cross-entropy training of this readout does not become optimization of the full belief free energy merely because its scores are divergences. The linear decoder used in the experiment of Section 7 is itself the unit-temperature Gibbs readout for the shared-covariance templates $p_v = \mathcal{N}(\Sigma_* w_v, \Sigma_*)$, with w_v the vth row of W_{out} and class prior $\log\omega_v = b_{\mathrm{out},v} + \kappa_{\mathrm{uni}}\log p_{\mathrm{uni}}(v) + \frac{1}{2}w_v^\top\Sigma_* w_v$ up to a constant, for any $\Sigma_* \succ 0$; the belief covariance cancels from every logit difference. The token-bank readout differs by giving each class its own template covariance.

Appendix E. Relation to a Normalized Latent-Source Model

A broader gauge-theoretic variational free energy (VFE) theory begins with a normalized generative law and a separately specified population recognition law. Geometry alone supplies neither. Let z collect the declared agent and auxiliary latent variables (the latent configuration, written y in the broader theory), let X denote fixed structural data, and let each record kernel have a normalized density ℓ_a. A normalized latent law p_θ^Z, constructed for example by finite directed kernel composition, gives

$$p_\theta(z, o \mid X) = p_\theta^Z(z \mid X)\prod_a \ell_a(o_a \mid z_{\partial a}, X). \tag{89}$$

Integrating the records first leaves the unit mass of p_θ^Z. Record scopes may overlap because each record is a conditional output of the same latent configuration. The factors are fixed while recognition varies. Latent variables may themselves take values in a space of probability laws; this does not permit a generative factor to read the recognition distribution being optimized.

Fix a regular observation slice with $0 < p_\theta(o \mid X) < \infty$. Let Q have density q, satisfy $Q \ll P_\theta(\cdot \mid o, X)$, and have finite expectations of $|\log q|$ and $|\log p_\theta(z, o \mid X)|$. Defining $\mathcal{L}_\theta(Q) = \mathbb{E}_Q[\log p_\theta(z, o \mid X) - \log q(z)]$ gives

$$\log p_\theta(o \mid X) = \mathcal{L}_\theta(Q) + D_{\mathrm{KL}}\big(Q\|P_\theta(\cdot \mid o, X)\big). \tag{90}$$

Indeed, Bayes' formula makes the log density ratio equal to $\log q - \log p_\theta(z, o \mid X) + \log p_\theta(o \mid X)$; integrating proves the identity. The KL is nonnegative and vanishes exactly at equality of the full laws. No factorization of Q is assumed. Outside the finite-density domain, the

extended free energy is defined as $-\log p_\theta(o \mid X) + D_{\mathrm{KL}}(Q \| P_\theta(\cdot \mid o, X))$, avoiding undefined differences of infinite expectations. This is the standard evidence-bound construction (Blei et al., 2017, Sec. 2.2, Eq. 14); consistent passive pushforwards preserve its relative entropy.

The broader theory also supplies a route to categorical attention. Give each receiver-head pair (i,h) an independent source label $J_i^{(h)}$ with prior $\pi_i^{(h)}$ and a binary compatibility record $\chi_i^{(h)} \in \{0,1\}$, and take

$$\begin{aligned} p_\theta(z, j, \chi, o \mid X) = p_\theta^Z(z \mid X) \prod_{i,h} \pi_i^{(h)}\big(j_i^{(h)}\big)\, \mathrm{Bern}\Big(\chi_i^{(h)}; \exp\big[-D^{(h)}_{i j_i^{(h)}}(z)/\tau_h\big]\Big) \\ \times \prod_a \ell_a(o_a \mid z_{\partial a}, X), \end{aligned} \tag{91}$$

in which no other factor reads a label. Each record kernel is normalized for every $\tau_h > 0$ because $D_{ij}^{(h)} \geq 0$; for a cost that is only finite, replacing D by $D - \min_k D_{ik}$ multiplies the selected-record likelihood by a label-independent factor. Conditioning on $\chi = \mathbf{1}$ gives the exact conditional rows $P(J_i^{(h)} = j \mid z, \chi, o) = \mathrm{softmax}_j\big(\log \pi_{ij}^{(h)} - D_{ij}^{(h)}(z)/\tau_h\big)$, independent across (i,h). This is the optimality-variable device of control as inference (Levine, 2018). Under $Q(dz, dj) = Q_Z(dz) \prod_{i,h} \beta_i^{(h)}(dj_i^{(h)})$ with $\mathbb{E}_{Q_Z} |D_{ij}^{(h)}| < \infty$, the row-dependent free energy and its optimizer are

$$\mathcal{F}_i^{(h)}(\beta_i^{(h)}) = D_{\mathrm{KL}}(\beta_i^{(h)} \| \pi_i^{(h)}) + \tau_h^{-1} \sum_j \beta_{ij}^{(h)} \mathbb{E}_{Q_Z} D_{ij}^{(h)}, \tag{92}$$

$$\beta_{ij}^{(h)*} = \mathrm{softmax}_j\big(\log \pi_{ij}^{(h)} - \mathbb{E}_{Q_Z} D_{ij}^{(h)}/\tau_h\big). \tag{93}$$

Expansion of the factorized density gives the first expression up to a row-independent term; a normalization multiplier gives the second. The pointwise generative posterior instead uses $D_{ij}^{(h)}(z)$ inside softmax. Multiplying a row objective by τ_h preserves its optimizer, but does not license independently rescaling sectors of one global evidence lower bound (ELBO).

The fixed-context Gaussian mixture of Section 5.1 supplies a different local joint with the same unit-temperature row objective: at unit temperature its KL is the projected Gaussian comparison cost plus categorical relative entropy. Appending the same normalized content law in the model and recognition factors makes that content law cancel, as in Eq. (54). It therefore permits the chosen message means without deriving the learned content maps or residual. The general Gaussian realization likewise permits singular conditional mean maps with positive-definite receiving noise, but that normalization fact does not select the predictive attention architecture.

The autoregressive softmax predictor already defines a normalized probability law, and its outer cross-entropy is the corresponding negative log-likelihood; in evaluation each conditional is computed from a bounded window, which leaves every factor normalized. For the attention sublayer, a normalized latent model exists in which routing is the exact label posterior and the content-plus-residual update is the exact predictive mean (Theorem 8). What remains unresolved is a latent joint and recognition construction that also covers the normalization and MLP maps, the collapse of the predictive mixture to a point state between layers, and the relation of cross-entropy training to that model's evidence. A model with no latent variables would make the evidence identity tautological without deriving those operations.

Appendix F. Additional Scope Results for Gaussian Predictive Architectures

The main text distinguishes geometric attention from a complete model of population inference and nonlinear state evolution. The following calculations state the population-covariance, Gaussian-closure, and causal-prediction conditions used in the main text. They concern the declared states and predictive maps.

F.1 Population covariance and link-residual costs

Let a joint population state have law $\mathcal{N}(\boldsymbol{\mu}, \Gamma)$, with covariance blocks $\Gamma_{ij} = \mathrm{Cov}(k_i, k_j)$ and $\Gamma_{ii} = \Sigma_i$. For a fixed link $L_{ij} : E_j \to E_i$, set $e_{ij} = k_i - L_{ij}k_j$ and $\bar{e}_{ij} = \mu_i - L_{ij}\mu_j$. Bilinearity of covariance gives

$$V_{ij} := \mathrm{Cov}(e_{ij}) = \Sigma_i + L_{ij}\Sigma_j L_{ij}^\top - \Gamma_{ij}L_{ij}^\top - L_{ij}\Gamma_{ji}. \tag{94}$$

This identity needs no inverse of L_{ij}. For any receiving covariance $R_i \succ 0$, expansion around the mean gives the expected quadratic cost

$$\mathbb{E}[e_{ij}^\top R_i^{-1} e_{ij}] = \bar{e}_{ij}^\top R_i^{-1} \bar{e}_{ij} + \mathrm{Tr}(R_i^{-1} V_{ij}). \tag{95}$$

The centered cross terms have zero expectation, leaving the trace term. These are Gaussian moment identities (Bishop, 2006); the two displayed moment formulas also hold for non-Gaussian laws with finite second moments.

A product approximation to the population sets the off-diagonal blocks Γ_{ij} to zero. Local state covariances alone do not justify that substitution. For a scalar example, take unit marginal variances, zero means, $L_{ij} = 1$, and cross covariance c, with $|c| < 1$. The expected squared residual is $2(1-c)$, while every individual Gaussian marginal is unchanged as c varies. Thus comparing local distributions and computing a correlated population residual are different operations.

A Gaussian record with mean $k_i - L_{ij}k_j$ and positive-definite noise can use Eq. (95) inside a normalized population model of Appendix E. Its expected log-likelihood cost is not generically the KL between the two projected Gaussian marginals in the main text. That KL has its own trace and determinant terms. Substituting a marginal divergence for the record cost changes the probabilistic model unless a separate identity establishes their equality.

F.2 Nonlinear mean maps and full Gaussian laws

An operation on a Gaussian mean does not automatically specify the pushforward of its entire law. To see this for LayerNorm (Ba et al., 2016), put $P_0 = I - \mathbf{1}\mathbf{1}^\top/K$ and define

$$\mathrm{LN}_\epsilon(x) = \frac{P_0 x}{\sqrt{\|P_0 x\|^2/K + \epsilon}}. \tag{96}$$

For $K = 2$, $\epsilon = 0$, and a nondegenerate Gaussian input, $x_1 = x_2$ has probability zero and

$$\mathrm{LN}_0(x) = \mathrm{sign}(x_1 - x_2)(1, -1). \tag{97}$$

The output law has two atoms. For $\epsilon > 0$ and arbitrary $K \geq 2$, the output remains in the proper zero-sum hyperplane, and it is bounded, $\|\mathrm{LN}_\epsilon(x)\|^2 < K$, so its law is not even a degenerate Gaussian. Fixed affine gains and biases map this support into an affine subspace of dimension at most $K-1$. Neither construction is a nondegenerate full Gaussian pushforward. Applying normalization only to μ_i, while retaining Σ_i for routing, remains a valid specified coordinate operation; it has a different statistical meaning.

F.3 Prefix-based causal prediction

A predictive conditional requires a state determined only by the available prefix and shared across all candidate next tokens. If one separately recomputes a binary state after inspecting each candidate target and assigns that target probability $3/4$, each row is normalized but the two candidate scores sum to $3/2$. They are not one conditional probability law. By contrast, a single prefix-based state followed by one categorical readout defines a normalized kernel, and the product of such kernels defines a finite sequence law by successive summation. The main architecture uses this direct causal construction. Outer backpropagation from training targets updates shared parameters; it does not insert those targets into the prediction-time state.

Appendix G. Experimental Record and Diagnostics

This appendix reserves the experimental record of the final runs. Each table and figure below is a placeholder to be replaced by recorded values or artwork.

PLACEHOLDER S1

Panel A: per layer, quantiles across tokens of the ratio $\|m_i\|/\|\mu_i\|$ of message to residual state, over training.

Panel B: common scale and scale-free invariants of the comparison maps (relative gains and principal angles of $S_{Q,h}, S_{K,h}$), and the largest singular value of $O_h T_h$, over training.

Panel C: mean and spread of the stored log-variances over training, with the curve implied by weight decay alone.

Panel D: counts, with denominators, of cost-bound activations, factorization fallbacks, frame-generator projections, and skipped updates.

TO COLLECT: per-table norms and invariants at every evaluation, and instrumented event counters.

Figure 6: Reserved training-dynamics and numerical diagnostics. The routing costs are unchanged when a common invertible map acts on $S_{Q,h}$ and $S_{K,h}$, and the messages are unchanged under $(T_h, O_h) \mapsto (BT_h, O_h B^{-1})$. Gradient flow conserves the corresponding quantities $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ and $T_h T_h^\top - O_h^\top O_h$, whereas adaptive optimizers without weight decay move along these directions; Panel B therefore separates motion along the symmetries from changes of the realized operator.

Reporting field	Final-run value or artifact identifier
Source and configuration	pending: source archive and hash, configuration schema, resolved operator selectors
Data and tokenizer	pending: dataset and split revision, tokenizer and version, vocabulary, concatenation policy, token-cache hashes, unique training tokens, passes
Architecture	pending: K, blocks, heads, ranks r_h, s_h, depth, normalization placement, residual, MLP width, decoder
Initialization	pending: prior means and covariances, frame generators, interface maps, as executed
Optimization	pending: learning rate and weight decay for every parameter table, schedule, warmup, clipping groups, post-update frame projection
Numerical realization	pending: precision per operation, cost bound, reduced-covariance factorization and failure policy, matrix-exponential policy
Training budget	pending: token presentations, updates, batch and accumulation, device-hours
Evaluation	pending: checkpoint-selection rule, context, stride, scored targets, masking

Table 6: Reproducibility contract for the final reported runs. Populate each field from the executing source archive and its saved metadata, and separate effective settings from accepted but inactive selectors.

Parameter role or resource	Evaluated run	Final run
Per-type tables (token frames, prior means and covariances, decoder)	pending	pending
Positional frame generators	pending	pending
Context-processing parameters (interface maps, MLP, normalization)	pending	pending
Parameters read per token	pending	pending
Optimizer state and peak device memory	pending	pending
Training throughput and device-hours	pending	pending
Inference latency at declared batch and context	pending	pending

Table 7: Reserved storage and cost record. Sparse access to per-type tables reduces the parameters read per token but not the stored parameters; report both, together with measured throughput under a declared precision.

Model arm	Seeds	Selected step	Validation CE	Test CE	Test PPL
pending	pending	pending	pending	pending	pending

Table 8: Reserved per-seed results for every arm of the main results table. List the model seed and the data-order seed, and apply the checkpoint-selection rule identically across arms.

Model and source	Tokenization	Context	Word-normalized test PPL
Evaluated realization	GPT-2 BPE	pending	pending
Published reference points	pending	pending	pending

Table 9: Reserved calibration of the headline result. Word-normalized perplexity is the exponential of the summed test negative log-likelihood divided by the number of word-level test tokens, after an audit confirming that both token streams encode the same text. Published rows use different protocols and are contextual references, not matched comparisons.

PLACEHOLDER S2

Panel A: per layer, the $H \times B$ matrix of Frobenius norms of each head's maps $S_{Q,h}, S_{K,h}, T_h, O_h$ restricted to each frame block.

Panel B: attention entropy per layer and head, with the minimum-entropy rows and the self-weights $\beta_{ii}^{(h)}$.

TO COLLECT: trained interface maps and validation-set routing weights.

Figure 7: Reserved head and frame-block diagnostics. Panel A measures whether communication heads coincide with the irreducible frame blocks of Appendix B or access several blocks through full-width maps.

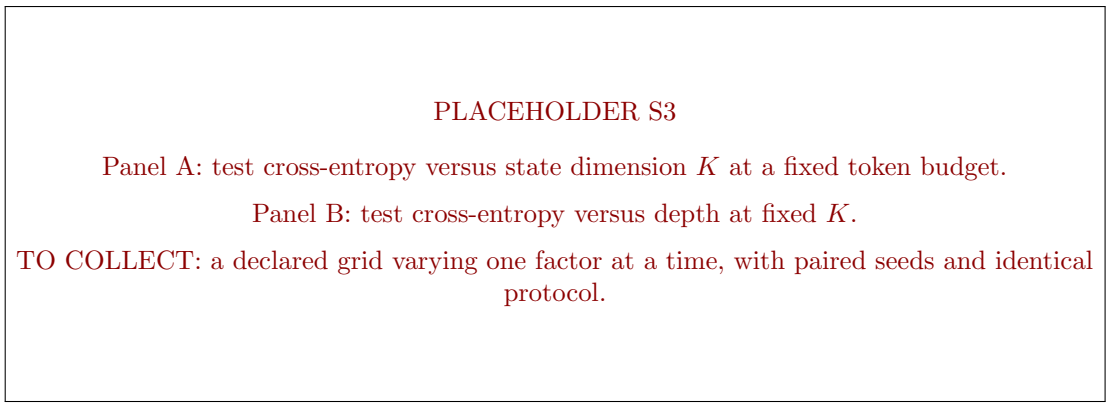


Figure 8: Reserved width and depth sweep. Existing preliminary runs differ in training duration, context, stride, learning rates, and covariance state, so they do not define this controlled comparison.

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